AP Chemistry Notes (Khan Academy)

Unit 1: Atomic Structure and Properties

1.1: Moles and Molar Mass

Mole: the unit of measurement in the International System of Units (SI) for the amount of substance

Molar mass: the mass in grams of one mole of a substance

Formula unit: the fundamental unit of an ionic compound

Avogadro's number = 6.022×10^{23}

- The number of particles in 1 mole (or mol) of a substance.
- Used to convert from moles to representative particles such as atoms, molecules, or formula units.

Element	Molar Mass (g/mol)
F	19.00
Sr	87.62

Calculate the number of formula units in a 33.8 g sample of strontium fluoride (SrF_2).

Calculate the molar mass of SrF_2:

1 mol Sr x 87.62 g/mol = 87.62 g Sr

2 mol F x 19.00 g/mol = 38.00 g F

87.62 + 38.00 = 125.62 g SrF_2

1 mole SrF_2 = 125.62 g = 6.022×10^{23} formula units

33.8 g SrF_2 x $\frac{6.022 \times 10^{23}\ formula\ units\ SrF_2}{125.62\ g}$
= 1.62×10^{23} formula units SrF_2

Element	Molar Mass (g/mol)
H	1.008
C	12.01
O	16.00

Calculate the number of moles in a 2.03 kg sample of citric acid ($C_6H_8O_7$).

Calculate the molar mass of $C_6H_8O_7$:

6 mol C x 12.01 g/mol = 72.06 g C

8 mol H x 1.008 g/mol = 8.064 g H

7 mol O x 16.00 g/mol = 112.0 g O

1 mole $C_6H_8O_7$ = 192.1 g

2.03 kg $C_6H_8O_7$ x $\frac{1000\ g}{1\ kg} \times \frac{1\ mol\ C_6H_8O_7}{192.1\ g\ C_6H_8O_7}$
= 10.6 mol $C_6H_8O_7$

1.2: Isotopes and Mass Spectrometry

Atomic mass unit (amu) or unified atomic mass unit (u): exactly 1/12 of a single neutral atom of carbon-12 (the most common isotope of carbon); equal to one gram/mole (the unit for molar mass).

Name	Charge	Mass (kg)	Mass (u)	Location
proton	1+	1.673×10^{-27}	1.007	inside nucleus
neutron	0	1.675×10^{-27}	1.009	inside nucleus
electron*	1-	9.109×10^{-31}	5.486×10^{-4}	outside nucleus

*The mass of an electron is so small relative to the masses of protons and neutrons that electrons are considered to have a negligible effect on the overall mass of an atom.

Isotopes: atoms with the same # of protons but different # of neutrons.

Isotopic notation/nuclear notation:

$$^{24}_{12}\mathrm{Mg}^{2+}$$

Element name

Mass number: (# protons) + (# neutrons)

Atomic number: (# protons)

Charge

Relative abundance of an isotope: the percentage of atoms with a specific atomic mass found in a naturally occurring sample of an element.

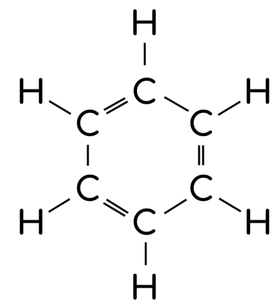
Average atomic mass: a weighted average calculated by multiplying the relative abundances of the element's isotopes by their atomic masses and then summing the products.

- $\sum_{i=1}^{n}$(relative abundance of isotopes x atomic mass)

Mass spectrometry: determines the relative abundance of each isotope.

- Ionizes atoms and molecules with a high-energy electron beam and then deflects the ions through a magnetic field inversely proportional to their mass-to-charge ratios (m/z)
- The mass spectrum of a sample shows the relative abundances of the ions on the y-axis and their m/z ratios on the x-axis. If z = 1 for all ions, then the x-axis can instead be expressed in units of atomic mass.

1.3: Elemental Composition of Pure Substances



Empirical formula: the simplest whole number ratio of elements in the compound. [CH]

Molecular formula: the kind and number of atoms of each element present in the molecular compound. [C_6H_6]

Structural formula: shows the structure of the compound. [→]

A 26.0 g sample of a compound contains 6.0 g of C, 1.0 g of H, and 19.0 g of F. What is the empirical formula of the compound?

Determine how many moles of each element are in the sample*:

6.0 g C x 1 mol/12.01 g ≈ 0.5 mol C

1.0 g H x 1 mol/1.008 g ≈ 1 mol H

19.0 g F x 1 mol/19.00 g ≈ 1 mol F

Find the simplest whole number ratio of atoms in the compound by dividing each of the mole values by the *smallest* of the values:

C: 0.5/0.5 = 1 C

H: 1/0.5 = 2 H

F: 1/0.5 = 2F

The empirical formula is CH_2F_2.

A pure sample of a compound contains 80% sulfur and 20% oxygen by mass. What is the empirical formula of the compound?

Determine the number of moles of each element in 100 g of the compound:

80 g S x 1 mol/32.06 g ≈ 2.5 mol S

20 g O x 1 mol/16.00 g ≈ 1.25 mol O

Find the simplest whole number ratio of atoms in the compound by dividing each of the mole values by the *smallest* of the values:

S: 2.5/1.25 = 2 S

O: 1.25/1.25 = 1 O

The empirical formula is S_2O.

A sample of a compound containing only carbon and hydrogen atoms is completely combusted, producing 11.0 g of CO_2 and 4.5 g of H_2O. What is the empirical formula of the compound?

When a compound containing only carbon and hydrogen atoms is completely combusted, all of the carbon atoms end up in the CO_2, and all of the hydrogen atoms end up in the H_2O.

Use the molar masses of CO_2 and H_2O to determine how many moles of carbon and hydrogen were in the sample of the compound before it was combusted*:

$$66.0\ g\ CO_2 \times \frac{1\ mol\ CO_2}{44.01\ g\ CO_2} \times \frac{1\ mol\ C}{1\ mol\ CO_2} \approx 1.5\ m \qquad 36\ g\ H_2O \times \frac{1\ mol\ H_2O}{18.02\ g\ H_2O} \times \frac{2\ mol\ H}{1\ mol\ H_2O} \approx 4\ m$$

Find the simplest whole number ratio of atoms in the compound:

C: 1.5 x 2 = 3 *H: 4 x 2 = 8*

The empirical formula of the compound is C_3H_8.

*The molar mass (g/mol) of any element is equal to its atomic mass (amu) on the periodic table.

1.4: Composition of Mixtures

A 1.05 g iron supplement contains 11.8% Fe by mass. The iron is present in the supplement as $C_4H_2FeO_4$ (s) (molar mass 169.90 g/mol). How many grams of $C_4H_2FeO_4$ (s) are in the iron supplement?

Use the mass percent of Fe to calculate the mass of Fe in the supplement:

1.05 g supplement x 0.118/1g supplement = 0.124 g Fe

Calculate how many moles of Fe and therefore $C_4H_2FeO_4$ are in the supplement:

$0.124 \text{ g Fe} \times \frac{1 \text{ mole Fe}}{55.85 \text{ g Fe}} \times \frac{1 \text{ mol } C_4H_2FeO_4}{1 \text{ mol Fe}}$
$= 2.22 \times 10^{-3} \text{ mol } C_4H_2FeO_4$

Use the molar mass of $C_4H_2FeO_4$ to convert from moles to grams:

$2.22 \times 10^{-3} \text{ mol } C_4H_2FeO_4 \times \frac{169.90 \text{ g } C_4H_2FeO_4}{1 \text{ mol } C_4H_2FeO_4} = 0.377 \text{ g } C_4H_2FeO_4$

A student determines that a 1.5 g mixture of $CaCO_3$ (s) and $NaHCO_3$ (s) contains 0.010 mol of $NaHCO_3$ (s). What is the mass percent of Na in the mixture*?

Determine the mass of Na in the mixture:

$0.010 \text{ mol } NaHCO_3 \times \frac{1 \text{ mol Na}}{1 \text{ mol } NaHCO_2} \times \frac{22.99 \text{ g Na}}{1 \text{ mol Na}} = 0.23 \text{ g Na}$

Find the mass percent of Na in the mixture:

0.23 g Na/1.5 g mixture x 100 = 15%

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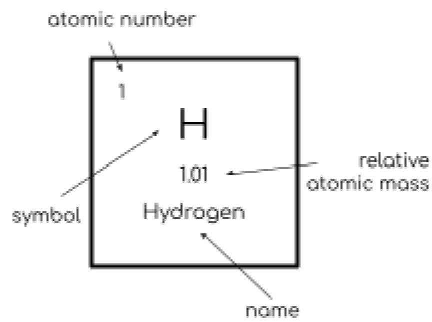
*The mass percent of a substance in a mixture can be determined by comparing the mass of the substance in the mixture to the total mass of the mixture.

1.5: Atomic Structure and Electron Configuration

Periodic table [link]:

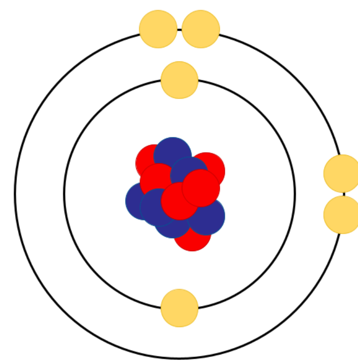
Groups (columns): organized by the number of valence electrons. Elements in a group have similar characteristics. (18 total)

Periods (rows): named by the number of electron/energy shells the element has. Arranged according to their mass. (7 total)



Bohr model: an atomic model showing the atom as a central nucleus containing protons and neutrons with the electrons in circular electron shells at specific distances. Each shell is assigned a number and the symbol n (the principal quantum number/main energy level of an electron).

Protons (+) | Neutrons (n/a) | Electrons (-)



Electron shell: an orbit followed by electrons around an atom's nucleus (can contain max $2n^2$ electrons in the nth electron shell) (determines reactivity)

Valence shell: the outermost electron shell

Valence electrons: electrons in the valence shell

Core electron: non-valence electron

In order to move between shells, an electron must absorb (move to a higher energy shell) or release (move to a lower-energy shell) an amount of energy corresponding exactly to the difference in energy between the shells.

Octet rule: most important elements in biology need 8 electrons in the valence shell to be stable. Some can be stable with an octet even with a 3n valence shell (18 electrons) **[s and p subshells are filled even though the entire shell is not]**. This rule also states that atoms with 1-4 electrons tend to lose them, and those with 5-7 electrons tend to gain electrons.

Group 1 (alkali metals): one electron in their outermost shells. Unstable as single atoms but can become stable by losing or sharing their one valence electron. [H, Li, Na]

Groups 3-12 are transition metals and mostly have two valence electrons. (chem.libretexts.org)

Group 17 (halogens): 7 electrons. Tend to achieve a stable octet by taking an electron from other atoms and becoming negatively charged ions. [F, Cl]

Group 18 (noble/inert gases): have outer electron shells that are full or satisfy the octet rule. Highly stable/non-reactive. [He, Ne, Ar]

Electron orbital: a three-dimensional description of the most likely location of an electron around an atom (where it spends 90% of its time). Can hold two electrons.

Subshell: sets of one or more orbitals. The first shell has one subshell, the second shell has two subshells, and so on. Designated by the letters *s* [1 spherical orbital], *p* [3 dumbbell-shaped orbitals at right angles to one another], *d* [5 orbitals], and *f* [7 orbitals].

The maximum number of electrons in a subshell is equal to 2(2l + 1), where the azimuthal quantum number (or orbital angular momentum quantum number) ℓ [pronounced 'ell'] is equal to 0, 1, 2, and 3 for s, p, d, and f subshells so that the maximum numbers of electrons are 2, 6, 10, and 14 respectively.

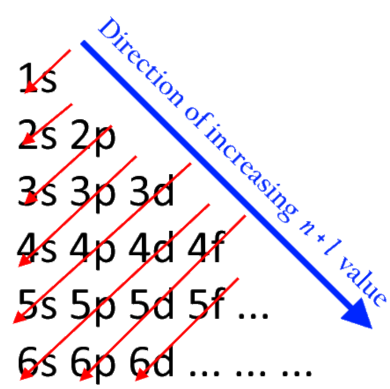
Helium (2 electrons) → completely fill the 1s orbital → $1s^2$

Lithium (3 electrons) → 2 fill the 1s orbital → 1 in the 2s orbital → $1s^2\ 2s^1$

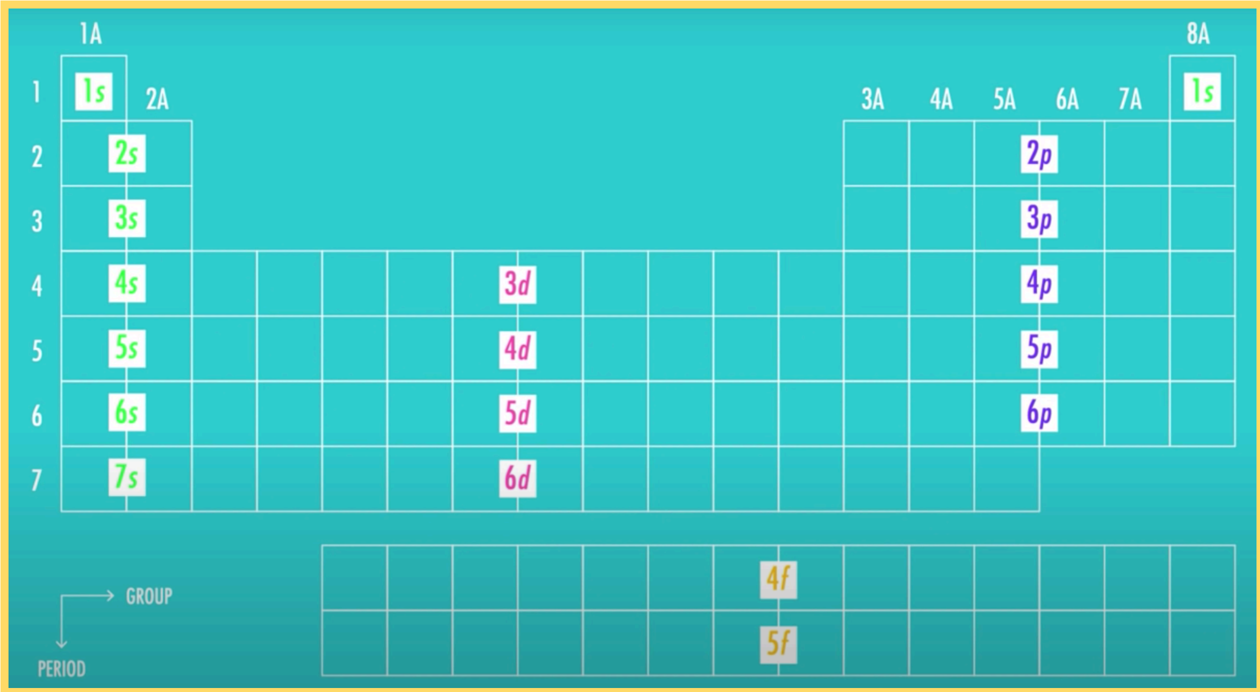
Neon (10 electrons) → 2 in 1s → 2 in each of the 2s and 2p orbitals [filled second shell] → $1s^2\ 2s^2\ 2p^6$

Aufbau [building] principle: in the ground state [state of lowest energy] of an atom/ion, electrons fill subshells from lowest to highest energy.

1. Electrons are assigned to subshells in order of increasing value of $n + \ell$.
2. For subshells with the same value of $n + \ell$, electrons are assigned first to the subshell with lower n.



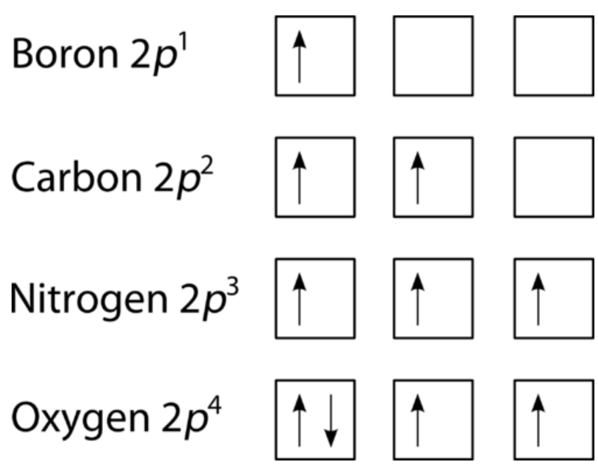
The states crossed by the same red arrow have the same $n + \ell$ value. The direction of the red arrow indicates the order of state filling. [Wikipedia]



[↑ the orbital being filled]

[s-block: groups 1 & 2 + He] [d-block: groups 3-12] [p-block: groups 13-18] [f-block: actinides and lanthanides]

Orbital filling diagram: a visual representation of the arrangement of all the electrons in a particular atom. Each orbital is shown as a square and the orbitals within a sublevel are drawn next to each other horizontally. Electrons are indicated by arrows inside the circles, with upwards arrows indicating one spin direction and downward arrows indicating the other direction.



1.6: Periodic Trends

[Mastering Periodic Trends | American Chemical Society]

[Effective Nuclear Charge - Electronic Structure - MCAT Content]

Anion: negatively charged ion

Cation: positively charged ion

Coulomb's law: the magnitude of the force between two charged particles is going to be proportional to the charge on the first particle times the charge on the second particle divided by the distance between those two particles squared

- $F \propto \frac{q_1 q_2}{r^2}$ OR $F_e = k\frac{q_1 q_2}{r^2}$
- k: Coulomb's constant. 8.99×10^9 N x m^2/C^2 [Newtons x $meters^2/Coulombs^2$]
- In the context of the periodic table, q_1 can be viewed as the effective positive charge from the protons in the nucleus of the atom and q_2 as the charge of an electron.
- Atomic/ionic radius: found by measuring the distances between atoms and ions in chemical compounds. Unit: pm (picometer) [one trillionth of a meter]

Effective charge: the net positive charge experienced by valence electrons [q_1 in Coulomb's law]. Approximated by the equation: $Z_{eff} = Z - S$, where Z is the atomic number and S is the number of shielding electrons.

- Shielding electrons: non-valence electrons
- Shielding effect/electron shielding/electron screening: the balance between the pull of the protons on the valence electrons and the repulsion forces from inner electrons. Explains why valence electrons are more easily removed from the atom.

(First) ionization energy/binding energy: the minimum energy required to remove electrons from a neutral version of that element. High when the Coulomb forces are high [high effective charge/z_{eff} and low radius].

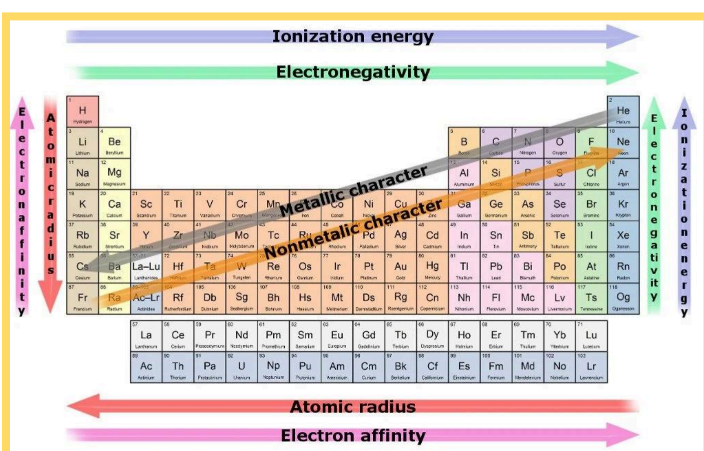
- Always positive because it takes energy to pull an electron away.
- Units: kilojoules/mole (kJ/mol)

- Decreases going down a group (easier to pull an electron away because of its distance from the nucleus)
- Increases going across a period (increase in z_{eff}/electrons pulled closer to the nucleus due to proton increase)
    - At some points, the amount of energy needed will drop slightly due to shielding provided by orbitals.
- Second ionization energy (the energy required to remove a second electron from a now 1^{+} ion of the element) is much greater than first ionization energy (energy required to remove the first electron) because it is closer to the nucleus.
- When electrons are removed in succession from an element, the transition from removing valence electrons to removing core electrons results in a large jump in ionization energy.
- Cations are smaller and anions are larger than the neutral atom.

Electron affinity: how much energy is released if an electron is added to a neutral version of a given element. High if Coulomb force is high [high effective charge/z_{eff} and low radius].

Electronegativity: the tendency for an atom of a given chemical element to attract shared electrons (or electron density) when forming a chemical bond (Wikipedia).

- Correlates strongly with electron affinity (if an element releases energy when it adds electrons, it is more likely to attract shared electrons)



Moving across periods means more protons added to the nucleus without any increase in electron shielding (new electrons are valence electrons, not core electrons). Ergo, the electrons are pulled closer to the nucleus and the atom is smaller.

The first two ionization energies for beryllium are shown below.

$Be(g) \rightarrow Be^{+}(g) + e^{-}$ $I_1 = 900$ kJ/mol

$Be^{+}(g) \rightarrow Be^{2+}(g) + e^{-}$ $I_1 = 900$ kJ/mol

Which of the following identifies the most probable value for the third ionization energy for Be and provides the best justification?

14,849 kJ/mol, because the electron removed during the third ionization is a core electron.

Beryllium has two valence electrons, so the electron removed from beryllium during the third ionization is a core electron. A large jump in ionization energy is typically associated with the transition from removing valence electrons to removing core electrons, so beryllium's third ionization energy is more likely to be 14,849 kJ/mol than 2672 kJ/mol.

Electron affinities generally become more favorable (more negative) as you move from left to right across a period. However, the electron affinity for P is *less* favorable than the electron affinity for Si, as shown below.

$Si(g) + e^{-} \rightarrow Si^{-}(g)$ EA = -134 kJ/mol

$P(g) + e^{-} \rightarrow P^{-}(g)$ EA = -72 kJ/mol

Which of the following best helps explain why the electron affinity for P is less favorable compared to the electron affinity for Si?

The additional electron in P^{-} experiences more electron-electron repulsion.

The electron added to phosphorus goes into a 3p orbital that already contains one electron, and the repulsion between these paired electrons causes P^{-} to be unstable. In contrast, the electron added to silicon goes into an empty 3p orbital, so these extra repulsions do not occur.

The first 4 ionization energies (kJ/mol) for elements X and Z are shown ↓.

	Element X	Element Z
First	900	738
Second	1757	1451
Third	14849	7733
Fourth	21007	10543

Which of the following correctly identifies elements X and Z, respectively?

Be and Mg

For both element X and element Z, the largest jump in ionization energy occurs when going from the second ionization energy to the third. This suggests that each element has two valence electrons.

Beryllium and magnesium each have two valence electrons. Magnesium is located below beryllium. First ionization energies generally decrease going down a group. Thus, magnesium must correspond to the element with the lower first ionization energy, and beryllium to the element with the higher first ionization energy.

Which of the following correctly identifies the element with the lower first ionization energy, Kr or Xe, and provides the best justification?

Xe, because of its larger atomic radius.

Xenon has more electron shells than krypton, so its atomic radius is larger. On average, the outermost electrons in larger atoms are farther from the nucleus, meaning they are held less tightly and require less energy to move.

Which of the following ground-state electron configurations represents the element with the lowest electronegativity?

[Kr] $4d^2 5s^2$

Electronegativity increases as you move from left to right across a period and decreases as you move down a group.

The least electronegative elements are found in the bottom left of the periodic table. In this case, the element closest to the bottom left of the table is Zr ([Kr] $4d^2 5s^2$).

Based on periodic trends and the data in the table below, which of the following is the most probable value for the ionic radius of K^+?

Element	Atomic radius	Ion	Ionic radius
Na	186 pm	Na^+	98 pm
K	227 pm	K^+	--

133 pm

The valence electrons in K^+ are in a lower electron shell than the valence electrons in K. Thus, K^+ must have a smaller radius than K.

K^+'s valence electrons are in a higher electron shell than Na^+'s.

Though K^+'s valence electrons are in the same electron shell as the valence electron in Na, K^+ has more protons in its nucleus and its electrons experience a greater effective nuclear charge. Thus, K^+ must have a smaller radius than Na.

Consider the following ions:

F^-, Na^+, O^{2-}, Mg^{2+}

Which of the following best predicts the trend in radii for these ions?

$Mg^{2+} < Na^+ < F^- < O^{2-}$

In this series, the ions are isoelectronic, meaning they have the same electron configuration.

The nuclear charge of each ion is equal to the number of protons in its nucleus, which is given by the atomic number.

The more protons in the nucleus, the stronger the attraction between the nucleus and the electron cloud. As a result, the ions with larger nuclear charges will have smaller radii.

1.7: Valence Electrons and Ionic Compounds

The first four ionization energies for element Z are shown below:

	Ionization energy (kJ/mol)
First	738
Second	1451
Third	7733
Fourth	10543

Which of the following is the most likely empirical formula of the ionic compound that forms between Z and nitrogen?

We need to know the charge of the ion that each element is most likely to form. When forming ions, elements typically gain or lose the *minimum* number of valence electrons necessary to achieve a full octet.

We can find out how many valence electrons Z has by looking for the large jump in ionization energy that typically occurs when transitioning from removing valence electrons to removing core electrons. For element Z, this jump occurs when going from the second ionization energy to the third. This suggests that Z has 2 valence electrons and will most likely lose 2 electrons to form an ion with a 2+ charge.

Nitrogen is in Group 5A and has 5 valence electrons. So, it is most likely to gain 3 electrons to form an ion with a 3- charge.

Ionic compounds are always electrically neutral, so the compound that forms between Z and nitrogen must contain 3 Z^{2+} ions for every two N^{3-} ions.

The most likely empirical formula is Z_3N_2.

-

The compound $CaCl_2$ is commonly used as a coagulant in the production of tofu from soy milk.

On the basis of periodic properties, which of the following compounds can most likely also be used in the production of tofu?

Elements in the same group of the periodic table often form analogous compounds with similar physical and chemical properties.

$CaCl_2$ contains Ca (Group 2) and Cl (Group 17). A compound formed by replacing either of these elements with an element from the same group is likely to yield a compound with similar coagulant properties as $CaCl_2$.

Of the answer choices, $MgCl_2$ is the only one formed by replacing one of the elements in $CaCl_2$ with an element from the same group (replacing Ca with Mg).

1.8: Photoelectron Spectroscopy

Photon: a particle representing a quantum of light or other electromagnetic radiation. Massless. (Oxford Languages)

Photoelectron spectroscopy (PES): an experimental technique that measures the relative energies of electrons in atoms and molecules.

- A sample is bombarded with high-energy radiation, which causes electrons to be ejected from the sample.
- The ejected electrons travel from the sample to an energy analyzer, where their kinetic energies are recorded, and then to a detector, which counts the number of photoelectrons at various kinetic energies.

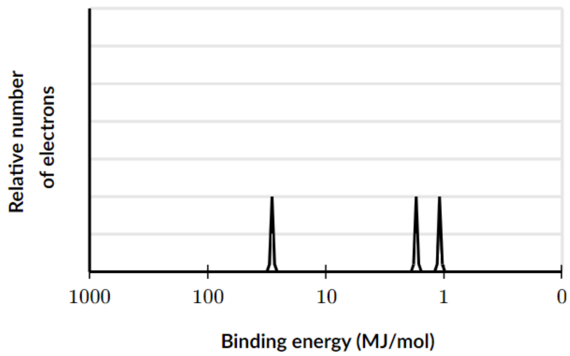
$BE = hv - KE_{electron}$: the equation for the ionization energy/binding energy (BE) needed to eject an electron from the sample.

- $KE_{electron}$: the kinetic energy of the photoelectron
- E_{photon}: energy of the photon
- $E_{photon} = hv$: h is Planck's constant (6.626×10^{-34} J · S) and v is the frequency of the photon in hertz
- $E_{photon} = BE + KE_{electron}$: the energy of the ionizing photon must be equal to the binding energy plus the kinetic energy of the photoelectron because of the law of conservation of energy
- Derived by combining $E_{photon} = hv$ and $E_{photon} = BE + KE_{electron}$ using the transitive property.

PES spectrum:

- The peaks in a PES spectrum correspond to electrons in different subshells of an atom. The peaks with the lowest binding energies correspond to valence electrons, while the peaks with higher binding energies correspond to core electrons.
- The binding energy of a peak shows the amount of energy required to remove the electron from its subshell.
- The intensity of the peak shows the relative number of electrons in the subshell.

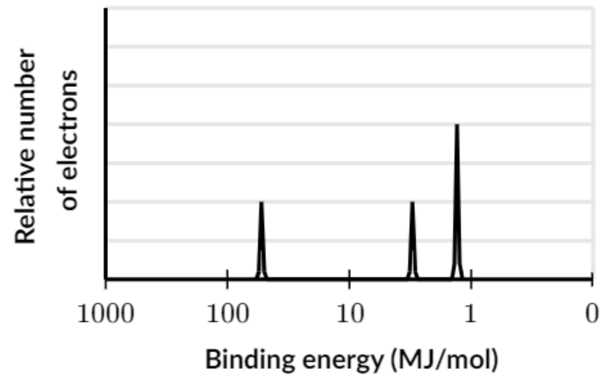
Q: The complete photoelectron spectrum* of C is shown below.



Based on the photoelectron spectrum of C, which of the following best predicts the spectrum of O?

*Idealized, less messy spectra with simplified data.

A:



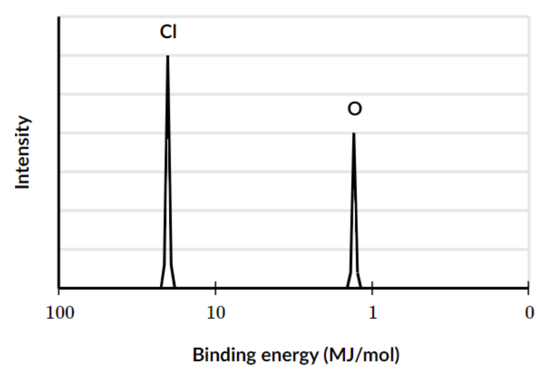
Oxygen has two more protons in its nucleus than carbon, so the electrons in oxygen experience a greater nuclear charge than the electrons in carbon. As a result, all of the peaks in the spectrum of oxygen should be shifted left (towards higher binding energies) relative to the peaks in the spectrum of carbon.

Oxygen has two more electrons in the 2p orbital, meaning that peak should be higher by two electrons.

The photoelectron spectra of the 2p electrons of O and Cl are shown to the right.

What best explains the difference in signal intensity for the two peaks?

Chlorine has more electrons in its 2p subshell than oxygen has.



In a photoelectron spectrum, peak intensity corresponds to the number of electrons in each subshell. According to their full electron configurations, chlorine ($1s^2 2s^2 2p^6 3s^2 3p^5$) has 6 electrons in its 2p subshell, while oxygen ($1s^2 2s^2 2p^4$) has only four. As a result, chlorine's 2p peak is one and a half times as intense as oxygen's 2p peak.

Unit 2: Molecular and Ionic Compound Structure and Properties

2.1: Types of Chemical Bonds

Ionic bonds: electrostatic forces of attraction between oppositely charged cations (+) and anions (-).

- Electrostatic force: the attractive or repulsive force between two electrically charged objects.
- In general, a bond has more ionic character when there is a *greater* electronegativity difference between the atoms in the bond.
- Generally form between metals and nonmetals.

Covalent bonds: formed between two atoms when both have similar tendencies to attract electrons to themselves (i.e. when both atoms have identical or fairly similar ionization energies and electron affinities).

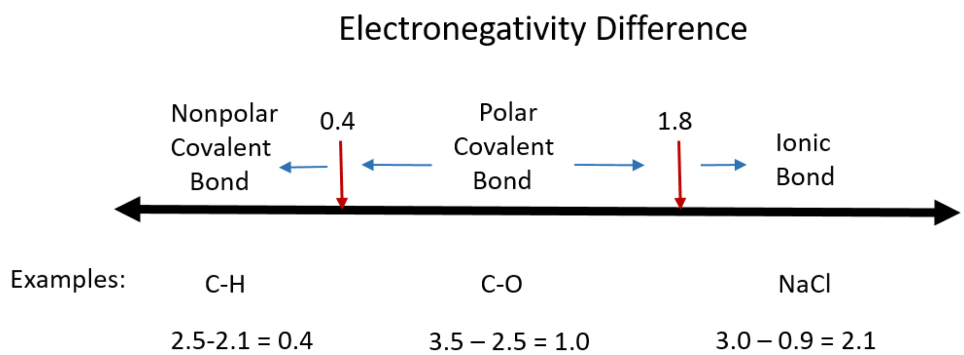
- Generally form between nonmetals.
- A covalent bond is represented by a line. Single bonds (one pair of electrons/2 electrons is shared by two atoms) have one line, double bonds (two pairs of electrons are shared by two atoms) have two, etc.

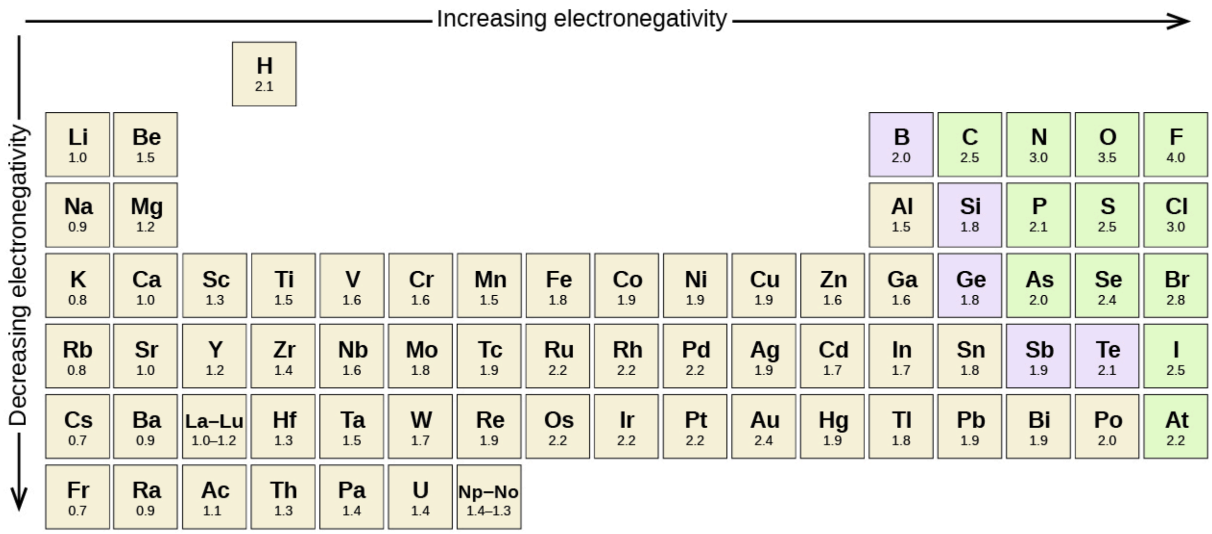
Nonpolar covalent bond (pure covalent bond): a covalent bond in which the bonding electrons are shared equally between the two atoms. The distribution of electrical charge is balanced between the two atoms.

Polar covalent bond (polar bond): a covalent bond in which the atoms have an unequal attraction for electrons, and so the sharing is unequal. The distribution of electrons around the molecule is no longer symmetrical.

- The polarity of a bond depends on the electronegativity difference between the atoms in the bond. The greater this difference, the more polar the bond.
- δ (lowercase delta): represents partial charge as the electron sharing is unequal. [δ^+ for partial positive charge and δ^- for partial negative charge]
- Ex: $^{\delta+}P - N^{\delta-}$ (Nitrogen is more electronegative and will have a partial negative charge. Phosphorus will thus have a partial positive charge.)

No chemical bond can be 100% ionic, and, except for those between identical atoms, 100% covalent bonds do not exist either.





[The electronegativity values derived by Pauling follow predictable periodic trends with the higher electronegativities toward the upper right of the periodic table (link)]

Metallic bond: the sharing of free electrons among a structure of cations. Metallic bonding accounts for many physical properties of metals, such as the ability to conduct electricity and heat well.

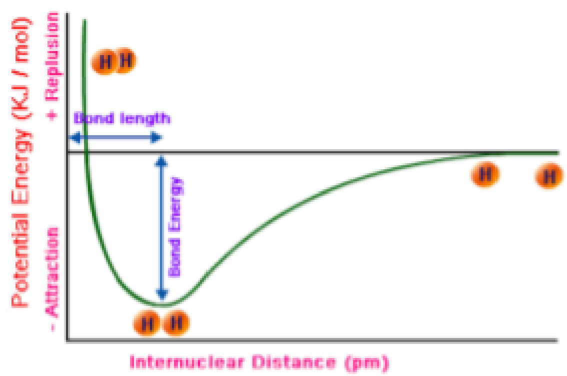
2.2: Intramolecular Force and Potential Energy

Diatomic: consisting of two atoms

Heteronuclear molecule: a molecule composed of atoms of more than one chemical element

Bond energy: the amount of energy needed to break the atoms involved in a covalent bond into free atoms.

- The length of a bond is influenced by both the bond order and the size of the atoms in the bond. The higher the bond order and the smaller the atoms, the shorter the bond.

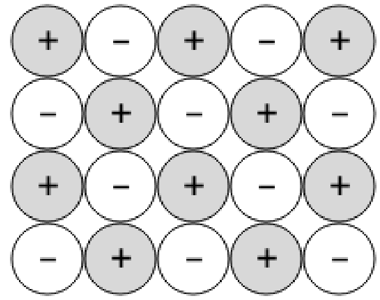


Crystal lattice: the symmetrical three-dimensional structural arrangement of atoms, ions, or molecules inside a crystalline solid as points (the geometrical arrangement of the atoms, ions, or molecules of the crystalline solid as points in space)

Lattice energy: the energy required to separate the ions in a crystal lattice into individual gaseous ions (energy required to break ionic bonds).

- Lattice energy depends on the force of attraction between cation and anion in the crystal lattice, which is given by Coulomb's law.

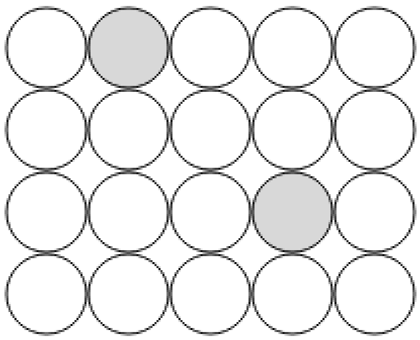
2.3: Structure of Ionic Solids



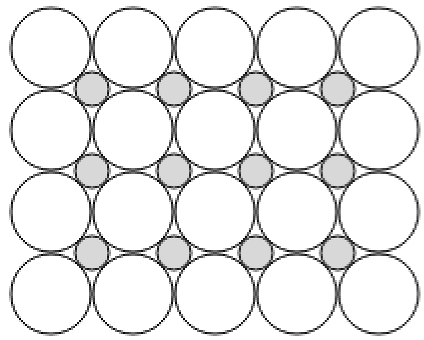
The cations and anions in an ionic solid are arranged in a lattice structure that maximizes the attractive forces between opposite charges and minimizes the repulsive forces between like charges. This is consistent with the structure depicted in this diagram, in which each ion is surrounded only by ions of the opposite charge.

2.4: Structure of Metals and Alloys

Substitutional alloy: forms between elements with similar atomic radii (within 15% of each other)



Interstitial alloy: forms between elements with significantly different atomic radii.



2.5: Lewis Diagrams

Central atom: usually the least electronegative atom.

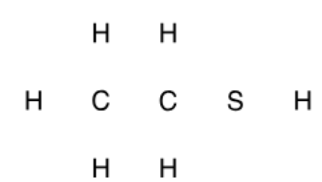
Terminal atom: an atom surrounding the central atom.

When drawing Lewis diagrams: if the central atom has an octet or exceeds an octet, you are usually done. If the central atom does not have an octet, create multiple bonds. Create bonds so the central atom will be as stable as possible.

Exceptions to the octet rule:

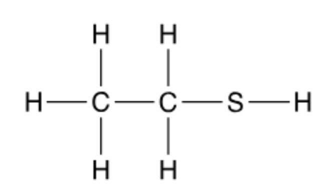
- Molecules with an odd number of electrons (ex: NO)
- Expanded-valence molecules: compounds with more than an octet of electrons around an atom (uses d orbitals) (3rd period and after).
- Molecules in which one or more atoms possess *less* than an octet of electrons (Boron, Beryllium).
- Duet rule (duplet rule): refers to the first five elements of the periodic table. They are most stable when the 1s orbital is filled with two (duet) electrons.

Ethanethiol, C_2H_6S, is a clear liquid with a strong odor. The compound is often added to otherwise odorless fuels such as natural gas to help warn of gas leaks. The skeletal structure of ethanethiol is shown →

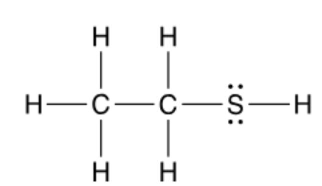


Correctly complete the Lewis diagram of ethanethiol:

1. Calculate the valence electrons: 2(4) + 6(1) + 1(6) = 20 electrons
2. Draw single bonds between the bound atoms in the molecule →



3. Distribute the remaining electrons throughout the structure. In this case, 4 electrons remain (20-16) after drawing the bonds. The duet and octet rules are already satisfied for the carbon and hydrogen atoms in the molecule. However, the sulfur atom has only 4 electrons. So, the final 4 electrons should be placed as lone pairs on the sulfur atom →

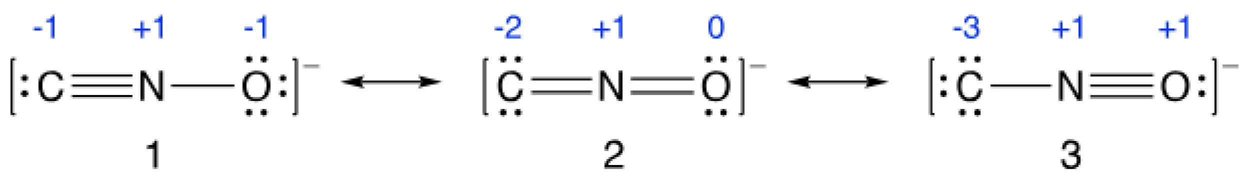


2.6: Resonance and Formal Charge

Resonance arises when more than one valid Lewis structure can be drawn for a molecule or ion. The overall electronic structure of the molecule or ion is given by the weighted average of these resonance structures and is referred to as the resonance hybrid.

Generally, a resonance structure contributes more to the resonance hybrid of a molecule when:

1) The formal charges of the atoms are minimized
2) Any negative formal charges are on more electronegative atoms and any positive formal charges are on more electropositive atoms.



In the above example, structure 3 contains the highest formal charges and likely contributes *less* to the overall structure than 1 or 2.

Structures 1 and 2 both contain negative formal charges totaling -2. However, structure 1 puts some of this charge on oxygen, which is more electronegative than carbon. This means that **structure 1 likely contributes *more* to the overall structure of CNO^- than structure 2.**

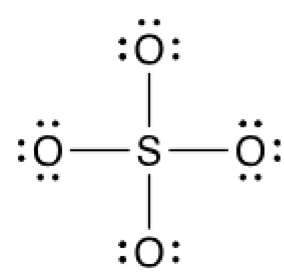
Formal charge: the charge assigned to an atom in a molecule.

- FC = (# of VE in free atom) - (# of VE in bonded atom)
- Where the # of VE in bonded atom = ½(# of shared electrons) + (# of lone electrons)

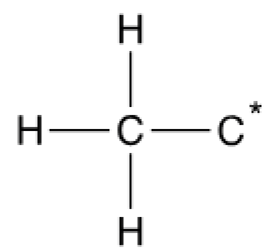
What is the formal charge of the sulfur atom in the diagram to the right?

FC = (VE on free S atom) - [½(shared electrons) + (lone pair electrons)]

FC = (6) - [½(8) + (0)] = 2

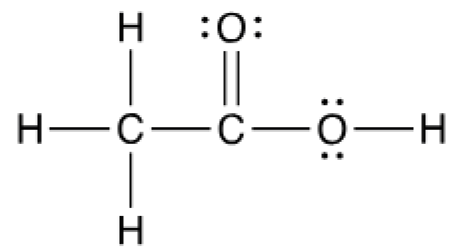


Acetic acid, CH_3COOH, is the main acidic component of vinegar. An incomplete Lewis diagram for the acetic acid molecule is shown to the right.



Which of the following Lewis diagrams best shows how the remaining atoms in acetic acid are arranged around the starred carbon?

In this Lewis diagram (→), all of the atoms have a full octet of electrons and a formal charge of zero. So, this diagram best represents the overall structure of acetic acid.



2.7: VSEPR

[9.2: The VSEPR Model - Chemistry LibreTexts]

[Geometry of Molecules - Chemistry LibreTexts]

[Dipole Moment - Definition, Detailed Explanation and Formula]

Valence shell electron-pair repulsion (VSEPR): a model used to predict the shapes of molecules and polyatomic ions. Based on the idea that the groups/clouds of electrons surrounding an atom will adopt an arrangement that minimizes the repulsions between them.

Dipole: the result of partial positive and negative charges in polar bonds.

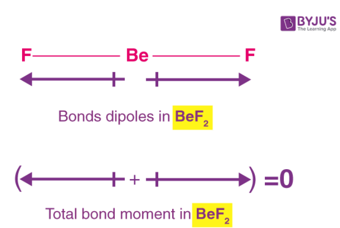
Dipole Moment (μ) = charge (Q) * distance of separation (r)

- A measure of the separation of negative and positive charges in a system.
- Represented by a slight variation of the arrow symbol. Denoted by a cross on the positive center and arrowhead on the negative center. This arrow symbolizes the shift of electron density in the molecule.

Bond dipole moment: a measure of the polarity of a chemical bond between two atoms in a molecule.

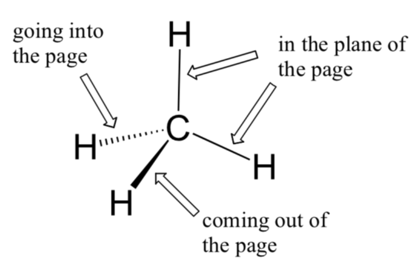
- $\mu = \delta$ (partial charge) * d (distance between partial charges)

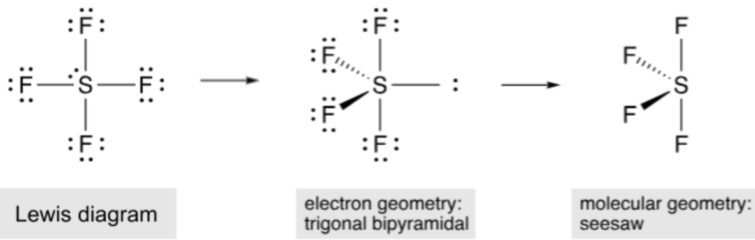
In a beryllium fluoride molecule, the bond angle between the two beryllium-fluorine bonds is 180°. Fluorine, being the more electronegative atom, shifts the electron density towards itself. The individual bond dipole moments in a BeF_2 molecule are illustrated to the right.



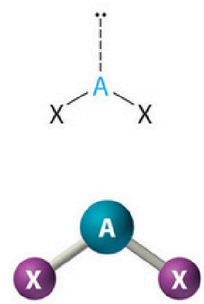
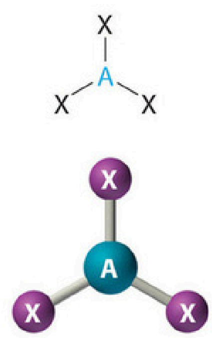
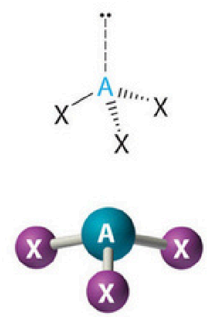
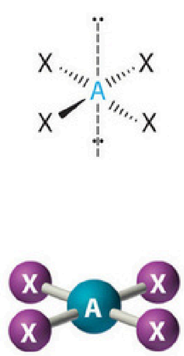
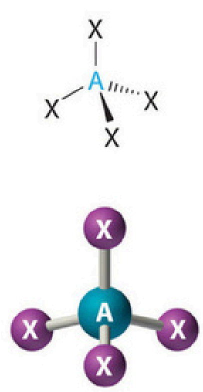
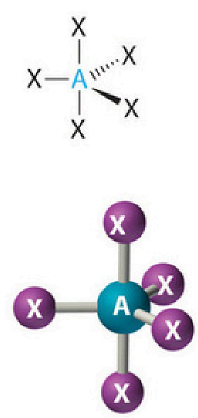
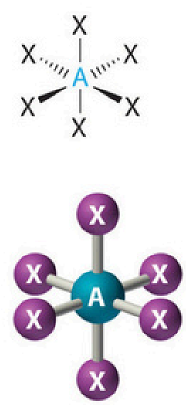
Molecular geometry: the arrangement of atoms in a molecule.

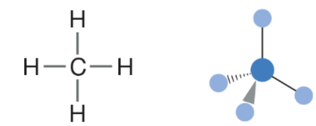
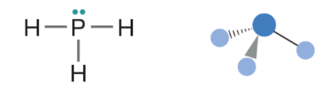
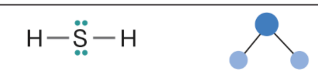
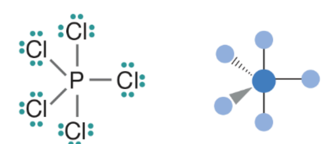
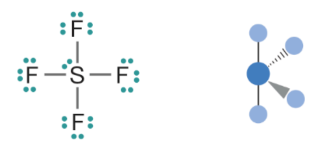
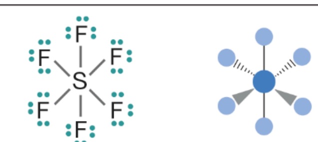
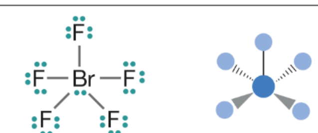
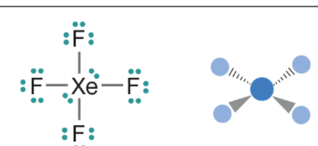
Electron geometry: the arrangement of electron pairs around a central atom





*To minimize repulsions, the lone pair will occupy one of the equatorial sites, resulting in a seesaw molecular geometry.

AX_mE_n Notation	AX_2	AX_2E	AX_3	AX_3E
Geometry	Linear 	Bent (V-shaped) 	Trigonal planar 	Trigonal pyramidal 
Idealized Bond Angles	180°	<180°	120°	<120°
AX_mE_n Notation	AX_4E_2	AX_4	AX_5	AX_6
Geometry	Square planar 	Tetrahedral 	Trigonal bipyramidal 	Octahedral 
Idealized Bond Angles	90°	109.5°	90°, 120°	90°

Electron Groups	Bonding Groups	Lone Pairs	Electron Geometry	Molecular Geometry	Approximate Bond Angles	Example
2	2	0	Linear	Linear	180°	
3	3	0	Trigonal planar	Trigonal planar	120°	
3	2	1	Trigonal planar	Bent	<120°	
4	4	0	Tetrahedral	Tetrahedral	109.5°	
4	3	1	Tetrahedral	Trigonal pyramidal	<109.5°	
4	2	2	Tetrahedral	Bent	<<109.5°	
5	5	0	Trigonal bipyramidal	Trigonal bipyramidal	120° (equatorial) 90° (axial)	
5	4	1	Trigonal bipyramidal	Seesaw	<120° (equatorial) <90° (axial)	
5	3	2	Trigonal bipyramidal	T-shaped	<90°	
5	2	3	Trigonal bipyramidal	Linear	180°	
6	6	0	Octahedral	Octahedral	90°	
6	5	1	Octahedral	Square pyramidal	<90°	
6	4	2	Octahedral	Square planar	90°	

[https://images.app.goo.gl/brjEZUAnHakNzXaQ9]

2.8: Bond Hybridization

[5.3: Hybridization of Atomic Orbitals - Chemistry LibreTexts]

[9.24: Sigma and Pi Bonds - Chemistry LibreTexts]

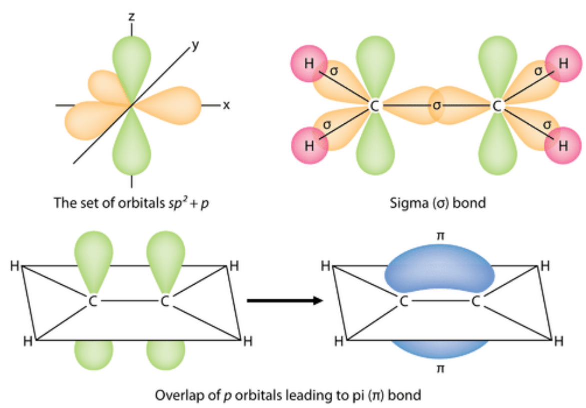
Angstrom (Å): a unit of length equal to 10^{-10} meters or 100 picometers.

Valence bond theory: assumes that all bonds are localized bonds formed between two atoms by the donation of an electron from each atom.

Orbital hybridization (hybridization): the concept of mixing atomic orbitals to form new hybrid orbitals (with different energies, shapes, etc., than the component atomic orbitals) suitable for the pairing of electrons to form chemical bonds in valence bond theory.

Sigma bonds (σ bonds): the strongest type of covalent bond. Formed by head-on overlapping between atomic orbitals.

Pi bonds (π bonds): a bond formed by the overlap of orbitals in a side-by-side fashion with the electron density concentrated above and below the plane of the nuclei of the bonding atoms.



In general, single bonds between atoms are always sigma bonds. Double bonds are comprised of one sigma and one pi bond. Triple bonds are comprised of one sigma bond and two pi bonds.

Steric number: the number of sigma bonds surrounding the atom plus the number of lone pairs on the atom. Used to determine the hybridization of an atom.

Steric #	Hybridization	Geometry
2	sp	linear (180°)
3	sp^2	planar (120°)
4	sp^3	tetrahedral (109.5°)

Unit 3: Intermolecular Forces and Properties

3.1: Intermolecular Forces

Intramolecular and intermolecular forces (article) | Khan Academy

Chemical bonds | Chemistry of life | Biology (article) | Khan Academy

Ion-Dipole Forces, Dipole-Dipole Forces

Isomer: each of two or more compounds with the same formula but a different arrangement of atoms in the molecule and different properties.

Ion-dipole force: an attractive force that results from the electrostatic attraction between an ion and a neutral molecule that has a dipole (cation attracts partially negative end of the polar molecule, anion attracts partially positive end).

Van der Waals forces: a general term for intermolecular interactions that do not involve covalent bonds or ions.

- Dipole-dipole interactions: attractive forces between the positive end of one polar molecule and the negative end of another polar molecule. Increase with increasing dipole moment.
- Hydrogen bond: a type of dipole-dipole interaction. Occurs between the lone pair of a highly electronegative atom (typically N, O, or F) and the hydrogen atom in an N-H, O-H, or F-H bond (partially positive hydrogen attracted to partially negative O, N, or F). Denoted by a dotted line. Strongest intermolecular force of attraction.
- London dispersion forces: weak attractions between molecules. Occur between atoms or molecules of any kind **when polar molecules induce, or create, a temporary dipole in otherwise nonpolar molecules**. Larger molar mass or surface area = stronger LDFs!

Intermolecular Force	Occurs between...	Relative strength
Dipole-dipole attraction	Partially oppositely charged ions	Strong
Hydrogen bonding	H atom and O, N, or F atom	Strongest of the dipole-dipole attractions
London dispersion attraction	Temporary or induced dipoles	Weakest

Which of the following compounds is least likely to be a gas at room temperature and pressure?

A) ~~Ethane, C_2H_6~~

B) ~~Formaldehyde, H_2CO~~

C) Methanol, CH_3OH

D) ~~Fluoromethane, CH_3F~~

A compound is less likely to be a gas at room temperature and pressure if it has a *higher* boiling point. A compound's boiling point is directly related to the strength of its intermolecular forces.

Ethane (C_2H_6): nonpolar, molecules only interact through dispersion forces.

Formaldehyde (H_2CO) and fluoromethane (CH_3F): polar, dipole-dipole forces and dispersion forces. Neither contains hydrogen bonds.

Methanol (CH_3OH): polar, contains an O-H bond. Hydrogen bonds, dipole-dipole and dispersion forces.

Methanol is the only one that can form hydrogen bonds, and hydrogen bonds are the strongest type of intermolecular force.

The molecular formulas and boiling points of three alcohols are given in the following table:

Name	Molecular Formula	Boiling Point (°C)
Methanol	CH_3OH	64.6
Ethanol	CH_3CH_2OH	78.3
1-Propanol	$CH_3CH_2CH_2OH$	97.2

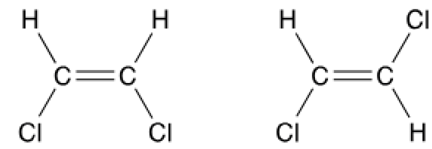
Which type of intermolecular force best accounts for the trend in boiling points seen above?

The boiling point of each alcohol is higher than that of the previous compound. A compound's boiling point is directly related to the strength of its intermolecular forces. The intermolecular force that best accounts for the trend must be the one that increases in strength down the series.

All of the alcohols are polar and contain an O-H bond, so their molecules interact through hydrogen bonding, dipole-dipole forces, and dispersion forces. Based on the structures of the three compounds, the hydrogen bonding and dipole-dipole forces are of similar strength.

However, because each alcohol has a larger molar mass than the one before it (meaning it has more electrons and is more polarizable), the dispersion forces should increase in strength going down the series.

The planar compound 1,2-dichloroethylene exists in two isomeric forms, which are shown to the right →



Isomer 1

Isomer 2

Based on these structures, which isomer of 1,2-dichloroethylene has the higher boiling point, and why?

A compound's boiling point depends on the strength of the intermolecular forces between its molecules. The stronger these forces, the more energy required for vaporization and the higher the boiling point.

Both isomers have dispersion forces of similar strength. However, isomer 1 is polar (since its C-Cl bond dipoles don't cancel), whereas isomer 2 is nonpolar (since its C-Cl bond dipoles *do* cancel). As a result, isomer 1 also has dipole-dipole forces, resulting in stronger intermolecular forces and a higher boiling point.

[A larger surface area will result in increased intermolecular contact and stronger dispersion forces.]

Based on the information in the table below, what explains why CBr_4 has a lower vapor pressure than CCl_4?

	Molar mass (g/mol)	Vapor pressure at 20° C (mm Hg)
CCl_4	153.8	115.0
CBr_4	331.6	0.8

A liquid's vapor pressure depends on the strength of the intermolecular forces between its molecules. The stronger these forces, the lower the rate of evaporation and the lower the vapor pressure.

CCl_4 and CBr_4 are both nonpolar, so the molecules only interact through dispersion forces. However, CBr_4 has a significantly larger molar mass, which means that its molecules have larger electron clouds and are more polarizable than those of CCl_4. As a result, the dispersion forces among molecules of CBr_4 are stronger, resulting in lower vapor pressure.

3.2: Properties of Solids

[AP Chem – 3.2 Properties of Solids | Fiveable]

[Covalent molecules - Periodicity - Higher Chemistry Revision - BBC Bitesize]

[Covalent Network Solids - Chemistry LibreTexts]

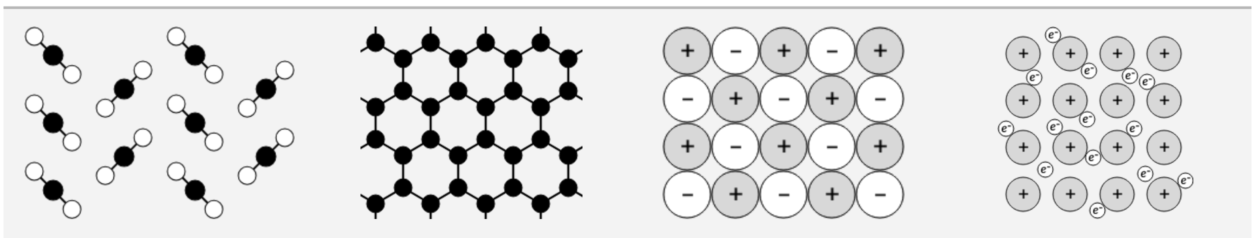
Discrete covalent molecules: small groups of atoms held together by strong covalent bonds inside the molecule and weak intermolecular forces between the molecules.

Sublime: transition directly from a solid to a gas

Localized electrons: the bonding electrons in chemical compounds.

Delocalized electrons: the nonbonding electrons in chemical compounds.

Molecular🧊	Covalent Network💎	Ionic🧂	Metallic✨



Molecular solids usually form from nonmetallic elements.

Covalent network solids include diamond, graphite, and silicon dioxide (SiO_2).

Ionic solids usually form between metallic and nonmetallic elements.

Metallic solids are composed of metallic elements.

Delocalized valence electrons allow for electricity to be conducted, high melting/boiling points, shine, and malleability/ductility in metals.

When ionic solids are melted or put into a solution and ions can move freely, then they can conduct electricity. In solid form, they are hard due to strong electrostatic attraction.

Molecular solids have strong covalent bonds (intramolecular forces) but weak intermolecular forces. The weak intermolecular forces cause it to be brittle, hard, and easy to break.

Type of Solid	Form of Unit Particle	Forces Between Particles	Properties
Molecular🧊	Atoms or molecules	LDFs, dipole-dipole, hydrogen bonding	*Fairly soft*, **low melting point**, bad conductor
Covalent Network💎	Atoms connected in a network of covalent bonds	Covalent bonds	*Very hard*, **very high melting point**, bad conductor
Ionic 🧂	Cations and anions	Electrostatic attractions	*Hard and brittle*, **high melting point**, bad conductor
Metallic✨	Atoms	Metallic bonds	***Varying hardness/melting points***, good conductor, malleable, ductile

[Electrostatic attractions can be evaluated using Coulomb's law, which tells us that the force of attraction between two ions is stronger when the charges on the ions are larger and when the distance between the ions is smaller.]

The melting point of SiO_2 is 1722° C, whereas the melting point of $SiCl_4$ is -69° C.

What best helps to explain why SiO_2 has a higher melting point than $SiCl_4$?

A solid's melting point is related to the attractive forces holding the solid together, which must be overcome in order for the solid to melt.

SiO_2 (s) is a common example of a covalent network solid. The covalent bonds in it are relatively strong and can only be overcome at high temperatures.

$SiCl_4$ (s) is a molecular solid. Its intermolecular forces are generally much weaker than covalent bonds and therefore can be overcome at lower temperatures.

So, SiO_2 has a higher melting point than $SiCl_4$ because the covalent bonds in SiO_2 (s) are stronger than the intermolecular forces in $SiCl_4$ (s).

3.3: Solids, Liquids, and Gases

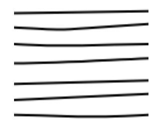
[Differences Between Amorphous & Crystalline Solids]

Polymer: a substance that has a molecular structure consisting chiefly or entirely of a large number of similar units bonded together. Has an amorphous structure when the chains that make up the polymer (represented as lines in the diagrams) are randomly arranged and intertwined.

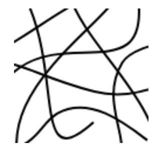
Crystalline: having a well-defined arrangement of atoms or molecules.

Amorphous: does not have a regular, repeating pattern of atoms or molecules, and therefore lacks a distinct crystalline structure.

Crystalline regions: contain polymer chains that are neatly folded or packed against one another. →



Amorphous regions: contain polymer chains that are randomly arranged and intertwined. →



Most polymeric solids are semicrystalline, meaning they contain both crystalline and amorphous regions. →

3.4: Ideal Gas Law

[What is the ideal gas law? (article) | Khan Academy]

[The Ideal Gas Law - Chemistry LibreTexts]

[Dalton's law of partial pressure (article) | Khan Academy]

[AP Chem – 3.4 Ideal Gas Law | Fiveable]

> Zeros to the right of the decimal place that are NOT merely placeholders are significant; these significant zeros will be to the right of non-zero significant digits. [When is a zero significant?]

Ideal gas:

1. Molecules do not attract or repel each other. Their only interactions are elastic collisions (collisions in which the amount of kinetic energy before and after remains the same).
2. The molecules themselves do not take up volume. The gas does take up volume since the molecules expand into a large region of space.

PV = nRT: [pressure] x [volume] = [number of moles of gas] x [ideal gas constant] x [temperature]

Units to Use for PV = nRT

$R = 8.314 \frac{J}{K \cdot mol}$	$R = 0.08206 \frac{L \cdot atm}{K \cdot mol}$	$R = 62.36 \frac{L \cdot Torr}{K \cdot mol}$
Pressure in kilopascals *kPa*	Pressure in atmospheres *atm*	Pressure in *Torr*
Volume in m^3 (1000 L)	Volume in liters *L*	Volume in liters *L*
Temperature in kelvin *K*	Temperature in kelvin *K*	Temperature in kelvin *K*

Bar: a unit of measurement equaling 100,000 pascals (100 kPa) not part of the International System of Units (SI) [R = 0.083145 with units K^{-1} mol^{-1}]

Standard Temperature and Pressure (STP): 0° C and 1 atm

K (kelvin) = °C + 273

Partial pressure: the pressure exerted by an individual gas in a mixture.

Dalton's law of partial pressures: the total pressure of a mixture of gases is equal to the sum of the partial pressures of the component gases.

- $P_{Total} = P_{gas\ 1} + P_{gas\ 2} + P_{gas\ 3} ...$
- $P_{gas\ 1} = x_1\ P_{total}$ where x is the mole fraction of gas 1 ($\frac{moles\ of\ gas\ 1}{total\ moles\ of\ gas}$)

Boyle's Law	Charles's Law	Gay-Lussac's Law
The relationship between the pressure and volume of an ideal gas under **constant temperature**.	The relationship between the volume and temperature of an ideal gas under **constant pressure**.	The relationship between the temperature and pressure of an ideal gas under **constant volume**.
$P_1V_1 = P_2V_2$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$

A sample of argon gas, Ar (g), is placed in a 3.80 container at 320 K. The gas pressure is 0.496 atm. How many moles of Argon gas are in the container? (Express in 3 significant figures.)

PV = nRT → $(0.496\ atm)(3.80\ L) = n \cdot \frac{0.082057\ L \cdot atm}{mol \cdot K} \cdot (320\ K)$ → n = 0.0718 mol

Under certain conditions, dimethyl ether decomposes according to the reaction represented above. A chemist runs the reaction at a constant temperature of 500° C in a rigid vessel. The $(CH_3)_2$ (g) is initially present in the vessel at a pressure of 0.60 atm.

What is the final pressure in the vessel after the reaction is complete?

The properties of gases are related by the ideal gas law, PV = nRT. The decomposition reaction is run at constant V and T, so the change in P will depend only on the change in n, or the number of moles of gas in the vessel. We can determine how n changes over the course of the reaction by examining the coefficients in the balanced chemical equation.

Three (3) moles of gas (1 mole of CH_4, 1 mole of CO, and 1 mole of H_2) are produced for every 1 mole of $(CH_3)_2O$ that decomposes. This means that n triples over the course of the reaction.

Since n is directly proportional to P, the pressure in the vessel also triples.

So, the final pressure in the vessel is 1.8 atm.

3.5: Kinetic Molecular Theory

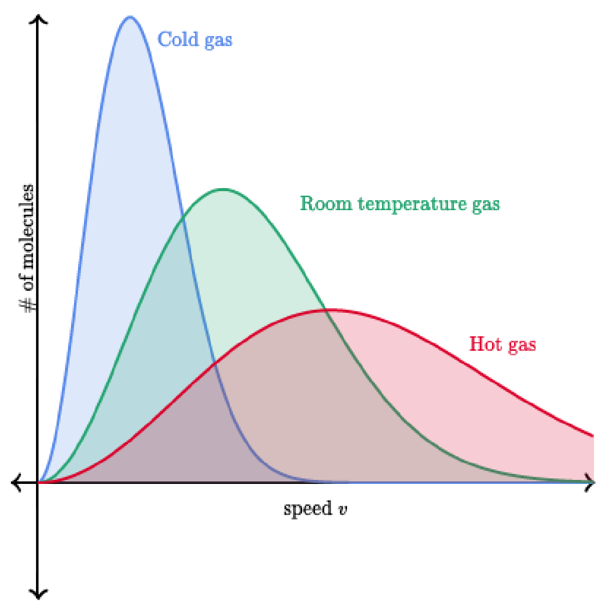
What is the Maxwell-Boltzmann distribution? (article) | Khan Academy

AP Chem – 3.5 Kinetic Molecular Theory | Fiveable

1. There are no attractive or repulsive forces between gas particles.
2. The particles of an ideal gas are separated by great distances compared to their size (gas particles have negligible, or no, volume because of how small and spread apart particles are).
3. Gas particles move in random, constant, straight-line motion.
4. Collisions are elastic: when gas particles collide, they transfer energy without a net loss - no energy is lost.
5. When observing particles, their kinetic energy is directly related to their velocity. All gases have the same average kinetic energy at a given temperature.
    - KE per molecule = ½ x mv^2: [Kinetic Energy per molecule] = ½ x [mass] x [velocity]^2

Maxwell-Boltzmann distribution: how the speeds of molecules are distributed for an ideal gas. The area under the curve is **100%**.

- $f(v) = \sqrt{\left[\frac{m}{2\pi kT}\right]^3} 4\pi v^2 e^{\left(-\frac{mv^2}{2kT}\right)}$



The higher the temperature, the higher the number of molecules reaching a higher speed.

Gas pressure depends on the frequency and intensity of the collisions between the gas particles and the container walls.

The rate of collisions with the container walls depends on the number of gas particles in the container and their average speed.

Molar mass (grams/mass) *does not* equal the number of moles!!!

3.6: Deviation from Ideal Gas Law

AP Chem – 3.6 Deviation from Ideal Gas Law | Fiveable

Non-ideal behavior of gases (article) | Khan Academy

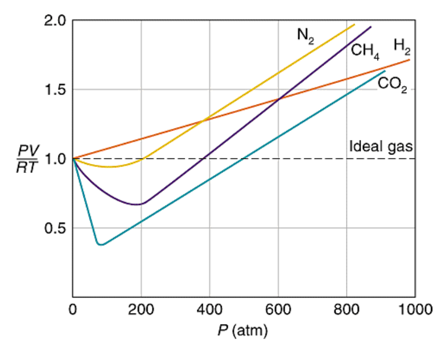
Gases can become attracted to each other.

When the gas particles are close together due to a large number of particles, this can cause more attractive forces. At low temperatures, gas particles move slower and spend more time around each other.

Polar molecules and larger molecules behave less ideally than smaller non-polar molecules. The IMFs (intermolecular forces) between polar molecules and larger molecules can cause these gas molecules to exert attractive forces on one another.

Gas particles can make up a significant portion of a gas sample's volume.

At high pressure, as shown through Boyle's Law, the volume of the container decreases. When volume decreases, the volume of gas particles begins to be more significant.



In real gasses, deviations from ideal pressures occur as a result of interparticle attractions and non-negligible particle volumes.

Negative pressure deviations result from interparticle attractions, which cause gas particles to strike the container walls less often than they would in the absence of attractive forces.

Positive deviations result from non-negligible [finite] particle volumes, which cause gas particles to strike the container walls more often than they would if the particle volumes were negligible.

3.7: Solutions and Mixtures

How to calculate molarity (article) | Khan Academy

Homogenous mixtures/solutions: mixtures with uniform composition.

Heterogeneous mixtures: mixtures with non-uniform composition.

Solvent: the chemical in the mixture that is present in the largest amount

Solutes: components other than the solvent

Molarity/molar concentration: the number of moles of solute per liter of solution. Unit: moles/liter, molar, M.

- $Molarity = \frac{mol\ solute}{L\ of\ solution}$
- Can be used to convert between the mass or moles of solute and the volume of the solution.
- Sometimes abbreviated by putting square brackets around the chemical formula of the solute (ex: $[Cl^-]$)

A 2.5 L solution contains 5.00 g of NaOH, which has a molecular weight of 40.00 g/mol. What is the molar concentration of sodium hydroxide in the solution?

mol NaOH
= 5.00 g x 1 mol/40.00 g
= 0.125 mol

Molar concentration
= mol solute/L solution
= 0.125 mol/2l50 L
= 0.0500 M

A 60.3 g sample of $Ba(OH)_2$ (molar mass 171.35 g/mol) is dissolved in enough water to produce 1.75 L of solution. What is the concentration of OH^- ions in the solution?

60.3 g $Ba(OH)_2$ x $\frac{1\ mol\ Ba(OH)_2}{171.35\ g\ Ba(OH)_2}$
= 0.352 mol $Ba(OH)_2$

Each mole of $Ba(OH)_2$ that dissolves dissociates into 1 mole of Ba^{2+} and 2 moles of OH^-.

0.352 mol $Ba(OH)_2$ x $\frac{2\ mol\ OH^-}{1\ mol\ Ba(OH)_2}$
= 0.704 mol OH^-

$M_{OH^-} = \frac{0.704\ mol\ OH^-}{1.75\ L\ soln}$ = 0.402 M

3.8: Representations of Solutions

AP Chem – 3.8 Representations of Solutions | Fiveable

Solvation: the process of a solvent dissolving a solute to form a solution.

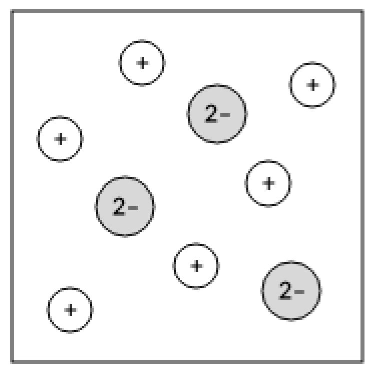
Aqueous (*aq*) solution: solutions in which water is the solvent.

Electrolyte: any substance whose aqueous solution contains ions.

- Strong electrolyte: a substance that completely dissolves (dissociates) in water into its constituent ions. Includes soluble salts, strong acids, and strong bases.
- Weak electrolyte: a substance that only partially dissociates in water and conducts electricity, but not as strongly as strong electrolytes. Includes weak acids and weak bases.

Nonelectrolyte: any substance whose aqueous solution does not contain ions.

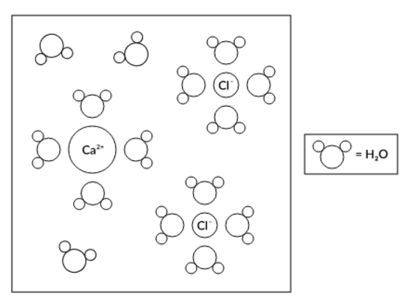
Which of the following diagrams best represents a solution of K_2CO_3 in water? (Water molecules not shown for simplicity.)



When an ionic solid such as K_2CO_3 dissolves in water, the solid dissociates into its component ions.

According to its chemical formula, K_2CO_3 contains two K^+ ions for every CO_3^{2-} ion. This means the correct diagram will show twice as many cations as anions.

A student was asked to draw a diagram showing the interactions of the different particles present in $CaCl_2$ (*aq*). What two errors are present in the student's diagram?



The diagram indicates that the Ca^{2+} ion is larger than the Cl^- ion, but the reverse is true.

All the water molecules surrounding the Cl^- ions are drawn in the wrong orientation.*

*When an ionic compound dissolves in water, the ions become surrounded by water molecules. In the case of anions, the water molecules orient themselves so that the positive (hydrogen) ends of the molecules point toward the negatively-charged ion.

3.9: Separation of Solutions and Mixtures Chromatography

AP Chem – 1.4 Composition of Mixtures | Fiveable

AP Chem – 3.9 Separation of Solutions and Mixtures Chromatography | Fiveable

AP Chem – 3.3 Solids, Liquids, and Gases | Fiveable

Principles of chromatography | Stationary phase (article) | Khan Academy

Chromatography: a technique that separates chemical species based on their interaction with a stationary phase.

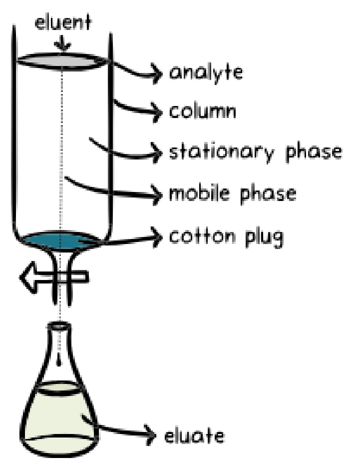
Mobile phase or carrier: solvent moving through the column.

Stationary phase or adsorbent: substance that stays fixed inside the column.

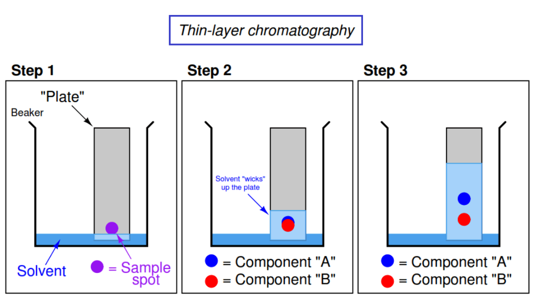
Eluent: fluid entering the column.

Eluate: fluid exiting the column (collected in flasks).

Analyte: mixture whose individual components have to be separated and analyzed.



Thin-Layer Chromatography (TLC): a technique that separates compounds in a sample using a stationary phase that is coated onto a thin layer of glass, plastic, or aluminum foil.



[Like dissolves like: polar substances dissolve in polar solvents and nonpolar substances dissolve in nonpolar solvents.]

1) ● = marker ink (easily separated by TLC)
2) The solvent climbs up the filter paper and begins to separate the sample spot by its substances.
3) Most TLC plates are made up of polar silica. Polar compounds are more strongly attracted to the stationary phase and will move less. Nonpolar compounds are more readily eluted with solvent.

Capillary action: the spontaneous rising of a liquid against gravity.

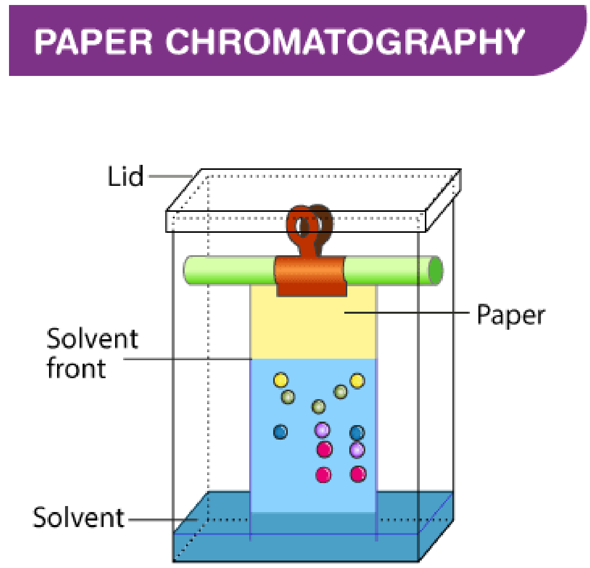
- Cohesive forces: forces between the liquid molecules that hold the liquid together.
- Adhesive forces: forces between liquid molecules and the container.

Paper chromatography: a small amount of the sample is applied to a strip of chromatography paper, which is then placed in a solvent.

Capillary action moves the solvent up the paper.

The more polar compounds in the sample have a greater affinity for the polar cellulose material that is the stationary phase and will move slower.

The nonpolar compounds have less affinity and move faster.



Column chromatography: a type of chromatography in which the stationary phase (typically silica or alumina) is packed into a column. As the compounds pass through the column, they interact with the stationary phase.

> TLC has several advantages over paper chromatography, including faster separation times, the ability to use multiple solvents, and the ability to visualize the separated compounds using a variety of techniques.
>
> Column chromatography is more complex and takes more time than the other two, but it allows for the separation of large quantities of a sample and can achieve higher purity of separated components.

Retention factor (R_f): the distance traveled by the individual component divided by the total distance traveled by the solvent. If both are polar, the higher the R_f value, the more polar the component.

- $R_f = \frac{distance\ from\ origin\ traveled\ by\ solute}{distance\ from\ origin\ traveled\ by\ solvent\ front}$

Principle of separation of different components: differential affinities (strength of adhesion) of the various components of the analyte towards the stationary and mobile phase result in differential separation of components.

- Affinity is dictated by two properties of the molecule: adsorption and solubility.
- Adsorption: the property of how well a component of the mixture sticks to the stationary phase.
- Solubility: the property of how well a component of the mixture dissolves the mobile phase.
- The higher the adsorption to the stationary phase, the slower the molecule will move through the column.
- Higher the solubility* in the mobile phase, the faster the molecule will move through the column.
- *Solubility is linked to polarity! [Like dissolves like!]

Filtration: uses a mesh/filter/porous barrier to separate solids from liquids in heterogeneous mixtures.

Distillation: separates components in liquid mixtures by evaporating the most volatile components first (the one with the lower boiling temperature).

	Simple distillation	Fractional distillation
Boiling points	Large difference in boiling points	Smaller difference in boiling points
Number of vaporization and condensation steps	One vaporization and condensation step	Multiple steps (allows for a more refined separation)
Purity of the separated compounds	Lower purity	Higher purity due to the additional steps

3.10: Solubility

A solute tends to be more soluble in a solvent when the intermolecular attractions between solute and solvent molecules are stronger.

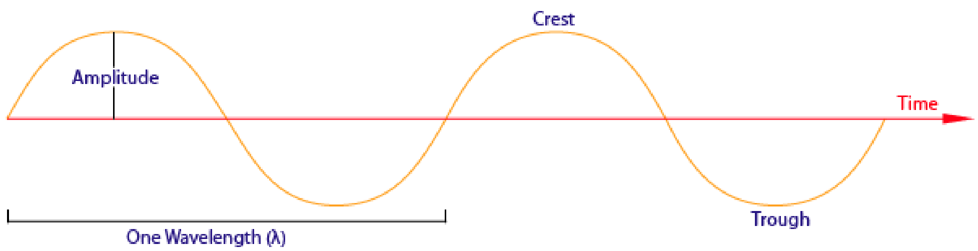
In general, nonpolar solvents are good at dissolving other nonpolar substances (like dissolves like) but not highly polar or ionic substances.

3.11: Spectroscopy and the Electromagnetic Spectrum

[Spectroscopy: Interaction of light and matter (article) | Khan Academy]

[AP Chem – 3.11 Spectroscopy and the Electromagnetic Spectrum | Fiveable]

[Light: Electromagnetic waves, the electromagnetic spectrum and photons (article) | Khan Academy]



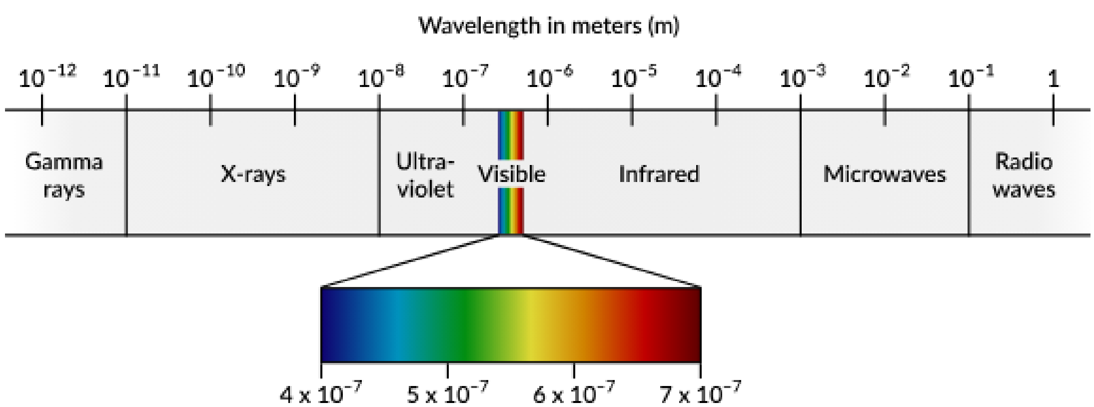
Trough: the lowest point of a wave.

Crest: the highest point of a wave.

Amplitude: the vertical distance between the tip of a crest and the wave's central axis. Associated with the brightness/intensity of a wave.

Wavelength: the horizontal distance between two consecutive troughs or crests.

Spectroscopy: the study of how different forms of electromagnetic radiation interact with atoms and molecules.



Infrared radiation: associated with molecular vibrational transitions.

Microwave radiation: associated with molecular rotational transitions.

UV/visible radiation: associated with electronic transitions.

3.12: Electronic Transitions in Spectroscopy

Photon: a particle representing a quantum of light or other electromagnetic radiation. Massless. (Oxford Languages)

Electronic transition: occurs when an electron either absorbs a photon and moves to a higher energy state or emits a photon and moves to a lower one. In both cases, **the energy of the photon is equal to the energy difference between the initial and final states.**

$E = hv$: energy is proportional to the frequency of the wave.

- E: the energy
- h: Planck's constant (6.626×10^{-34} J · S)
- v: the frequency

$c = \lambda v$: frequency and wavelength are inversely related.

- c: the speed of light (3×10^{8} m/sec)
- λ (the Greek lambda): the wavelength (in meters)

$E = hc/\lambda$: the energy of a photon is inversely proportional to its wavelength.

Nanometer (nm): equivalent to 10^{-9} of a meter

The first ionization energy of rubidium is 4.03×10^{5} J/mol. Calculate the longest wavelength, in nanometers, of electromagnetic radiation that is capable of ionizing an atom of Rb.

Calculate the amount of energy required to ionize a Rb *atom* (not a mole!) by dividing by Avogadro's number:

- $\frac{4.03\times10^{5}\,J}{1\,mol\,Rb} \times \frac{1\,mol\,Rb}{6.022\times10^{23}\,atoms\,Rb} = 6.69 \times 10^{-19}\,J\,per\,atom\,Rb$

Rearrange the equation relating the energy of a photo to wavelength and substitute in the values:

- $\lambda = \frac{hc}{E} \rightarrow \frac{(6.626\times10^{-34}\,J\cdot S)(2.998\times10^{8}\,m/s}{(6.69\times10^{-19}\,J)} = 2.97 \times 10^{-7}$

Convert from meters to nanometers:

- $2.97 \times 10^{-7}\,m \times \frac{1\,nm}{10^{-9}} = 297\,nm$

3.13: Beer-Lambert Law

[Spectroscopy: Interaction of light and matter (article) | Khan Academy]

Spectrophotometer: a device that measures the absorbance of the solution.

Adsorption: the measure (0-1) of a solution's opacity.

- An abundance of 0 means light totally passes through the solution (the solution is completely clear).
- An abundance of 1 means that no light passes through the solution (the solution is completely opaque).
- When a solution becomes darker, it means it is absorbing more visible light.

The Beer-Lambert Law ($A = \epsilon lc$)*: relates absorbance to the concentration of colored species in solution.

- A: the adsorption of the solution (unitless).
- ϵ (lowercase epsilon): the molar absorptivity constant (a constant unique to each compound, given in units of $M^{-1}\ cm^{-1}$).
- l (sometimes referred to as b): the path length of the solution container (in cm).
- c: the concentration of the solution in molarity.

Unit 4: Chemical Reactions

4.1: Net Ionic Equations

How to Write Net Ionic Equations | ChemTalk

Molecular equation/balanced equation: ionic compounds or acids are represented as neutral compounds using their chemical formulas.

- $AgNO_3$ (aq) + NaCl (aq) → AgCl (s) + $NaNO_3$ (aq)

Complete ionic equation: a molecular equation that separates the molecules into their ion forms.

- Ag^+ (aq) + NO_3^- (aq) + Na^+ (aq) + Cl^- (aq) → AgCl (s) + Na^+ (aq) + NO_3^- (aq)

Spectator ions: ionic species that remain unchanged during a reaction.

- NO_3^-(*aq*) and Na^+(*aq*)

Soluble ionic compounds, strong acids, and strong bases should be separated into their constituent ions in a complete ionic equation. Insoluble salts and weak acids should remain as one unit.

- AgCl (s) is insoluble in water.

Net ionic equation: a molecular equation showing only the species involved in the chemical reaction.

- Ag^+(*aq*) + Cl^-(*aq*) → AgCl (s)

Aqueous $SrCl_2$ reacts with aqueous Li_3PO_4 to produce a solid precipitate of $Sr_3(PO_4)^2$ and aqueous LiCl. What is the balanced net ionic equation for this reaction?

Write the balanced molecular equation:

$3SrCl_2$ (aq) + $2Li_3PO_4$ (aq) → $Sr_3(PO_4)_2$ (s) + 6LiCl (aq)

Rewrite this equation as a complete ionic equation (ionic compounds dissolved in water are present as *separate* ions):

$3Sr^{2+}$ (aq) + $6Cl^-$ (aq) + $6Li^+$ (aq) + $2PO_4^{3-}$ (aq) → $Sr_3(PO_4)_2$ (s) + $6Li^+$ (aq) + $6Cl^-$ (aq)

Identify and remove the spectator ions:

$3Sr^{2+}$ (aq) + $2PO_4^{3-}$ (aq) → $Sr_3(PO_4)_2$ (s)

$Cr_2(SO_4)_3$ (aq) + $3(NH_4)_2CO_3$ (aq) → $Cr_2(CO_3)_3$ (s) + $3(NH_4)_2SO_4$ (aq)

Identify the spectator ions in the reaction above.

Write the complete ionic equation:

$2Cr^{3+}$ (aq) + $3SO_4^{2-}$ (aq) + $6NH_4^+$ (aq) + $3CO_3^{2-}$ (aq) → $Cr_2(CO_3)_3$ (s) + $6NH_4^+$ (aq) + $3SO_4^{2-}$ (aq)

Identify which ions remain unchanged on both sides of the equation:

$2Cr^{3+}$ (aq) + $3SO_4^{2-}$ (aq) + $6NH_4^+$ (aq) + $3CO_3^{2-}$ (aq) → $Cr_2(CO_3)_3$ (s) + $6NH_4^+$ (aq) + $3SO_4^{2-}$ (aq)

The final net ionic equation: **$2Cr^{3+}$ (aq) + + $3CO_3^{2-}$ (aq) → $Cr_2(CO_3)_3$ (s)**

Ions That Form *Soluble* Compounds	Exceptions
Group 1 ions (Li^+, Na^+, etc.)	
Ammonium (NH_4^+)	
Nitrate (NO_3^-)	
Acetate ($C_2H_3O_2^-$ or CH_3COO^-)	
Chlorate (ClO_3^-)	
Halides (Cl^-, Br^-, I^-)	When combined with Ag^+, Pb^{2+}, or Hg_2^{2+}
Sulfates (SO_4^{2-})	When combined with Ag^+, Ca^{2+}, Ba^{2+}, or Pb^{2+}

Ions that form insoluble compounds **except when combined with Group 1 ions or ammonium (NH_4^+)**:

- Carbonate (CO_3^{2-})
- Chromate (CrO_4^{2-})
- Phosphate (PO_4^{3-})
- Sulfide (S^{2-})
- Hydroxide (OH^-) **[also insoluble when combined with Ca^{2+}, Ba^{2+}, Sr^{2+}]**

4.2: Representations of Reactions

The solutions of $Pb(NO_3)^2$ (aq) and Na_2S (aq) represented (↓) are mixed, causing a precipitate to form. The balanced equation for the reaction is shown below:

$Pb(NO_3)_2$ (aq) + Na_2S (aq) → PbS (s) + 2 $NaNO_3$ (aq)

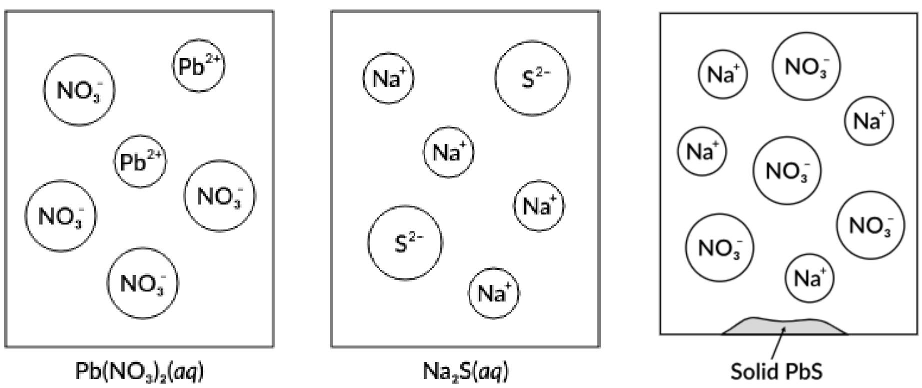
Which of the following diagrams best represents the mixture after the reaction has proceeded as completely as possible?

The identity of the precipitate formed is PbS (s).

PbS (s) contains one Pb^{2+} ion for every S^{2-} ion.

Since the diagrams of the reactant solutions show there are equal numbers of Pb^{2+} and S^{2-} ions before the reaction, this means there will not be any Pb^{2+} or S^{2-} ions remaining in the solution after the reaction.

So, the only ions that will be present in the solution after the reaction are the spectator ions, Na^+ and NO_3^-.



4.3: Physical and Chemical Changes

Types of Chemical Reactions | ChemTalk

In general, chemical processes (chemical reactions) involve changes in intramolecular forces (chemical bonds), while physical processes involve changes only in intermolecular forces (such as dispersion forces or hydrogen bonding).

Basic types of chemical reactions:

- Synthesis: a reaction that occurs when two atoms interact to form one atom. [A + B → AB]
- Decomposition: a reaction that occurs when a compound breaks down into two or more atoms. [AB → A + B]
- Single replacement (single displacement): a reaction that occurs when a new compound is formed when one element is substituted for another element in a compound, creating a new element and a new compound as products. [AB + C → B + AC]
- Double replacement (double displacement): a reaction in which the cationic or the anionic species switch places, creating two new products. [AB + CD → AC + BD]

4.4: Stoichiometry

Stoichiometry: the branch of chemistry that deals with the relationship between the relative quantities of substances taking part in a chemical reaction.

Given the following reaction:
$2KClO_3 \rightarrow 2KCl + 3O_2$

How many moles of KCL will be produced from 15.0 g of $KCLO_3$?

$\frac{15.0\ g}{122.55\ g/mol}$ = 0.122 moles of $KClO_3$

The mole ratio of $KCLO_3$/KCl in the reaction is 2/2.

x = 0.122 moles of KCl produced.

Given the following reaction:
$4Fe + 3O_2 \rightarrow 2Fe_2O_3$

How many moles of Fe_2O_3 will be produced from 18.0 g of Fe, assuming O_2 is available in excess?

$\frac{18.0\ g}{55.85\ g/mol}$ = 0.322 moles of Fe

The mole ratio of Fe/Fe_2O_3 in the reaction is 4/2.

Fe/Fe_2O_3 = 4/2 = 0.322 moles/x.

x = 0.161 moles of Fe_2O_3 produced.

Balance the following chemical equation:

▯H_2SO_4 + ▯$Pb(OH)_4 \rightarrow$ ▯$Pb(SO_4)^2$ + ▯H_2O

1) Treat the sulfate polyatomic ion (SO_4) as an atom, symbolized by X:

   $H_2SO_4 + Pb(OH)_4 \rightarrow Pb(SO_4)_2 + H_2O$

   $H_2X + Pb(OH)_4 \rightarrow PbX_2 + H_2O$

2) There is 1 X on the left and 2 X on the right, so multiply H_2X by 2.

   $2H_2X + Pb(OH)_4 \rightarrow PbX_2 + H_2O$

3) That gives us 8 H on the left and only 2 on the right, so multiply H_2O by 4.

   $2H_2X + Pb(OH)_4 \rightarrow PbX_2 + 4H_2O$

Limiting reactant (limiting reagent): a reactant in a chemical reaction that can limit the amounts of products formed by the reaction.

Theoretical yield: the amount of product formed when the limiting reactant is fully consumed in a reaction (determined by stoichiometry).

Given the following reaction: $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$

How many grams of $MgCl_2$ will be produced from 16.0 g of $Mg(OH)_2$ and 11.0 g of HCl?

$\frac{16.0\ g}{58.32\ g/mol}$ = 0.274 moles of $Mg(OH)_2$

$\frac{11.0\ g}{36.46\ g/mol}$ = 0.302 moles of HCl

The mole ratio of $Mg(OH)_2$/HCl in the reaction is ½.

$Mg(OH)_2$/HCl = ½ = x/0.302 moles

x = 0.151 moles of $Mg(OH)_2$ needed. We have 0.274 moles of $Mg(OH)_2$, which is more than we need. Therefore, HCl is the limiting reagent.

The mole ratio of HCl/$MgCl_2$ in the reaction is 2/1.

HCl/$MgCl_2$ = 2/1 = 0.302 moles/x

x = 0.151 moles of $MgCl_2$ produced.

0.151 moles $MgCl_2$ x 95.20 g/mole = 14.4 grams of MgCl.

$(CH_3)_2O\ (g) \rightarrow CH_4\ (g) + CO\ (g) + H_2\ (g)$

4.5: Oxidation-Reduction (Redox) Reactions

Difference Between Oxidation Number and Charge

Oxidation-reduction (redox) reaction: a reaction that involves the transfer of electrons between chemical species (the atoms, ions, or molecules involved in the reaction).

OIL RIG: Oxidation Is Loss, Reduction Is Gain (of electrons)

Oxidation number (oxidation state): the imaginary charge an atom would have if all of the bonds to the atom were completely ionic (electrons completely pulled away).

- **An atom of a free element** (a chemical element that is not combined with or chemically bonded to other elements) has an oxidation number of 0.
- A **monatomic ion** (an ion consisting of exactly one atom) has an oxidation number equal to its charge.
- When combined with other elements, **alkali metals (Group 1A)** always have an oxidation number of +1, while **alkaline earth metals (Group 2A)** always have an oxidation number of +2.
- **Fluorine** has an oxidation number of -1 in all compounds.
- **Hydrogen** has an oxidation number of +1 in most compounds. The major exception is when hydrogen is combined with metals, as in NaH or $LiAlH_4$. In these cases, the oxidation number of hydrogen is -1.
- **Oxygen** has an oxidation number of -2 in most compounds. The major exception is in peroxides (two oxygen atoms connected by a single covalent bond), where oxygen has an oxidation number of -1.
- **The other halogens (Cl, Br, and I)** have an oxidation number of -1 in compounds unless combined with oxygen and fluorine. For example, the oxidation number of Cl in the ion ClO_4^- is +7 (since O has an oxidation number of -2 and the overall charge on the ion is -1).
- **The sum of the oxidation numbers** for all the atoms in a neutral compound is equal to 0, while the sum for all atoms in a polyatomic ion is equal to the charge on the ion.
- **Pure elements** have an oxidation number of 0 due to their nonpolar covalent bonds (electrons are shared equally between the atoms).

	Oxidation Number	Charge
Definition	The charge of the central atom of a coordination compound when all the bonds around this atom are ionic bonds.	The net electrical charge of an atom.
Determination	Determined considering the number of electrons that are either removed or gained by that atom.	Determined considering the total number of electrons and protons in the atom.
Value	The same chemical element can have different oxidation numbers depending on the atoms surrounding it.	The charge of an atom depends on the number of electrons and protons in the atom.
Denotation	Using Roman numerals inside brackets*	Using Hindu-Arabic numbers with either plus (+) or minus symbols (-)*

*Oxidation numbers are written with the sign *before* the number. Ions are written with the charge *after* the number.

A positive oxidation number indicates the loss of electrons, while a negative oxidation number indicates the gain of electrons.

Half-reaction method of balancing:

1) Divide the equation into two half-reactions, one representing oxidation and one representing reduction.
2) Balance the equations for mass and charge. (If necessary, adjust so that the number of electrons transferred in each equation is the same.)
3) Add the half-reaction equations together for the balanced overall equation.

Oxidation half-reaction: shows the reactants and products participating in the oxidation process.

Reduction half-reaction: shows the reactants and products participating in the reduction process.

Balance this equation: Co^{3+}(*aq*) + Ni(*s*) → Co^{2+}(*aq*) + Ni^{2+}(*aq*)

This equation is balanced with respect to mass, but not for charge. The net charge on the left side is 3+ and 4+ on the right side.

Oxidation: Ni (*s*) → Ni^{2+} (aq)

Like the overall equation, our half-reaction is balanced for mass but not charge. Balance it out by adding two electrons to the right side so the net charge on each side is 0:

Oxidation: Ni (*s*) → Ni^{2+} (aq) + $2e^-$

Reduction: Co^{3+} (aq) → Co^{2+} (aq)

Include an electron on the left side of the equation for charge balance:

Co^{3+} (aq) + e^- → Co^{2+} (aq)

The electrons for reduction come from the oxidation half-reaction.

The oxidation half-reaction involves the transfer of *two* electrons, while the reduction half-reaction involves the transfer of only *one* electron.

The electrons won't cancel out if the equations are added together now.

So, multiply the reduction half-reaction by 2 first:

- 2[Co^{3+} (aq) + e^- → Co^{2+} (aq)]
- $2Co^{3+}$ (aq) + $2e^-$ → 2 Co^{2+} (aq)

Add the two half-reactions together and cancel out the electrons on both sides.

$$
\begin{array}{rl}
 & \text{Ni (s)} \rightarrow \text{Ni}^{2+}\text{(aq)} + \cancel{2e^-} \\
+ & 2\text{Co}^{3+}\text{(aq)} + \cancel{2e^-} \rightarrow 2\text{Co}^{2+}\text{(aq)} \\
\hline
 & \text{Ni (s)} + 2\text{Co}^{3-}\text{(aq)} \rightarrow \text{Ni}^{2+}\text{(aq)} + 2\text{Co}^{2+}\text{(aq)}
\end{array}
$$

Use H^+ ions and H_2O molecules to balance equations of reactions occurring in acidic solutions.

Use OH^- ions and H_2O molecules to balance equations of reactions occurring in basic solutions.

A substance is oxidized when it gains oxygen or loses hydrogen.

A substance is reduced when it loses oxygen or gains hydrogen.

Oxidizing agents oxidize other substances and get reduced.

Reducing agents reduce other substances and get oxidized.

Balance the equation for the reaction of copper metal with nitrate ion in the acidic solution: Cu (s) + NO_3^- (aq) → Cu^{2+} (aq) + NO_2 (g)

Divide the equation into half-reactions:

- Oxidation: Cu (s) → Cu^{2+} (aq) + $2e^-$
- Reduction: NO_3^- (aq) → NO_2 (g)

Balance the O atoms in the reduction half-reaction by adding $1H_2O$ to the right side:

- NO_3^- (aq) → NO_2 (g) + H_2O (l)

Balance the H atoms by adding H^+ ions to the left side:

- NO_3^- (aq) + $2H^+$ (aq) → NO_2 (g) + H_2O (l)

Balance the equation for charge:

- Reduction: NO_3^- (aq) + $2H^+$ (aq) + e^- → NO_2 (g) + H_2O (l)

Equalize number of electrons transferred:

- 2[NO_3^- (aq) + $2H^+$ (aq) + e^- → NO_2 (g) + H_2O (l)]
- $2NO_3^-$ (aq) + $4H^+$(aq) + $2e^-$ → $2NO_2$ (g) + $2H_2O$(l)

Combine the two half-reactions and cancel out the electrons:

- Cu (s) + 2 NO_3^- (aq) + 4 H^+ (aq) → Cu^{2+} (aq) + $2NO_2$ (g) + $2H_2O$ (l)

Balance the equation for the reaction of permanganate and iodide ions in basic solution: MnO_4^- (aq) + I^- (aq) → MnO_2 (s) + I_2 (aq)

Divide the equation into half-reactions:

- Oxidation: I^- (*aq*) → I_2 (*aq*)
- Reduction: MnO_4^- (aq) → MnO_2 (s)

Balance the oxidation reaction for mass and charge:

- Oxidation: $2I^-$ (aq) → I_2 (aq) + $2e^-$

Balance the reduction reaction for mass and charge:

- First balance the half-reaction as if it occurs in acidic solution for a quicker approach:
  $MnO_4^-(aq) + 4H^+(aq) \rightarrow MnO_2(s) + 2H_2O(l)$
- To account for the fact that the half-reaction actually takes place in basic solution, add OH^- to both sides to neutralize the H^+:
  $MnO_4^-(aq) + 4H^+(aq) + 4OH^-(aq) \rightarrow MnO_2(s) + 2H_2O(l) + 4OH^-(aq)$
- The H^+ and OH^- ions on the left side form H_2O molecules:
  $MnO_4^-(aq) + 4H_2O(l) \rightarrow MnO_2(s) + 2H_2O(l) + 4OH^-(aq)$
- Some of the H_2O molecules cancel out:
  $MnO_4^-(aq) + 2H_2O(l) \rightarrow MnO_2(s) + 4OH^-(aq)$
- Balance the half-reaction for charge:
  $MnO_4^-(aq) + 2H_2O(l) + 3e^- \rightarrow MnO_2(s) + 4OH^-(aq)$

Equalize the number of electrons transferred:

- $3[2I^-(aq) \rightarrow I_2(aq) + 2e^-] = 6I^-(aq) \rightarrow 3I_2(aq) + 6e^-$
- $2[MnO_4^-(aq) + 2H_2O(l) + 3e^- \rightarrow MnO_2(s) + 4OH^-(aq)]$
  $= 2MnO_4^-(aq) + 4H_2O(l) + 6e^- \rightarrow 2MnO_2(s) + 8OH^-(aq)$

Add the half-reactions together:

- $2MnO_4^-(aq) + 6I^-(aq) + 4H_2O(l) \rightarrow 2MnO_2(s) + 3I_2(aq) + 8OH^-(aq)$

4.6: Introduction to Acid-Base Reactions

[Brønsted-Lowry acids and bases (article) | Khan Academy]

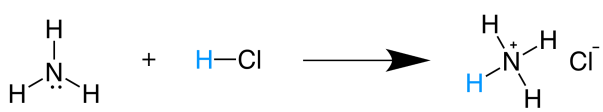
Brønsted-Lowry theory: describes acid-base interactions in terms of proton transfer between chemical species.

Brønsted-Lowry acid: any species that can donate a proton (H^+).

- H^+ is a proton because it (H) has lost its only electron.
- H^+ takes the form of H_2O^+ in aqueous solutions.

Brønsted-Lowry base: any species that can accept a proton. Must have at least one lone pair of electrons to form a new bond with a proton.

$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$



- HCl donates its proton (the Brønsted-Lowry acid)
- NH_3 has a lone pair that accepts the proton (the Brønsted-Lowry base)

Water is known as an amphoteric or amphiprotic substance, meaning it can act as either a Brønsted-Lowry acid or a Brønsted-Lowry base.

Strong acid: a species that dissociates completely into its constituent ions in an aqueous solution.

Name	Formula
Hydrochloric acid	HCl
Hydrobromic acid	HBr
Hydroiodic acid	HI
Sulfuric acid	H_2SO_4
Nitric acid	HNO_3
Perchloric acid	$HClO_4$

Weak acid: a species that does *not* dissociate completely.

Strong base: a base that ionizes completely in an aqueous solution.

- Common strong bases include Group 1 and Group 2 hydroxides (OH).

Weak base: a base that does *not* ionize completely in an aqueous solution.

- Common weak bases include neutral nitrogen-containing compounds (ammonia, trimethylamine, pyridine, etc.)

Hydrogen phosphate, HPO_4^{2-}, can act as a weak base *or* as a weak acid in an aqueous solution.

What is the balanced equation for the reaction of hydrogen phosphate acting as a weak base in water?

Since hydrogen phosphate is acting as a Brønsted-Lowry base, water must be acting as a Brønsted-Lowry acid. The addition of a proton to hydrogen phosphate results in the formation of $H_2PO_4^-$, dihydrogen phosphate:

- HPO_4^{2-} (aq) + H^+ (aq) → $H_2PO_4^-$ (aq)

Since hydrogen phosphate is acting as a *weak* base in this particular example, we will need to use equilibrium arrows [$\rightleftharpoons$]* to show that the reaction is reversible:

- HPO_4^{2-} (aq) + H^+ (aq) $\rightleftharpoons$ $H_2PO_4^-$ (aq)

*Equilibrium arrows are used with weak acids or weak bases, as they do not dissociate completely.

Conjugate acid: the species formed after the base *accepts* a proton.

Conjugate base: the species formed after an acid *donates* its proton.

HCl (aq) + H_2O (l) → H_3O^+ (aq) + Cl^- (aq) [acid + base → acid + base]

In this reaction, HCl donates a proton to water, acting as a Brønsted-Lowry acid. After HCl donates its proton, the Cl^- ion is formed; thus, Cl^- is the conjugate base of HCl.

> Conjugate pair 1 = HCl and Cl^-

Because water accepts a proton from HCl, water is acting as a Brønsted-Lowry base. When water accepts a proton, H_3O^+ is formed. Therefore, H_3O^+ is the conjugate acid of H_2O.

> Conjugate pair 2 = H_2O and H_3O^+

Each conjugate acid-base pair contains one Brønsted-Lowry acid and one Brønsted-Lowry base; the acid and base differ by a single proton.

It will generally be true that a reaction between a Brønsted-Lowry acid and base will contain two conjugate acid-base pairs.

4.7: Introduction to Titration

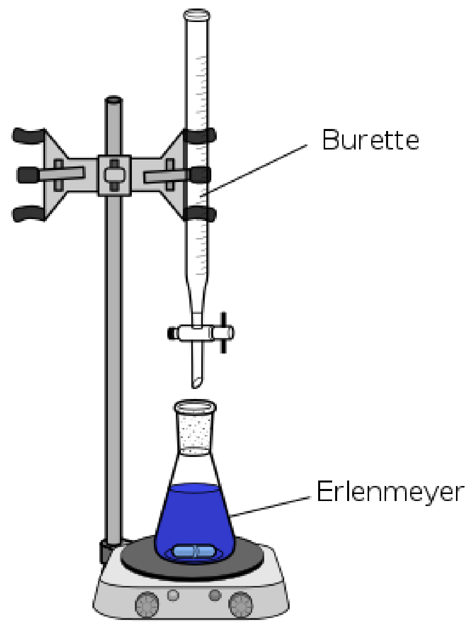
AP Chem – 4.6 Introduction to Titration | Fiveable

Titration curves & equivalence point (article) | Khan Academy

Titration: an experimental method used to determine the unknown concentration of a chemical solution.

Titrant: a solution of known concentration used to determine the concentration of the unknown solution in a titration.

- Usually added to the unknown solution using a burette (buret), a long, narrow tube with a stopcock used for volume precision.



Analyte (titrand): the unknown solution whose concentration is determined in a titration.

- Typically in the Erlenmeyer flask below the burette.

Equivalence point: The titrant-analyte mole ratio is the same in the balanced equation and the reaction. The reactants are completely consumed. Indicates the end of the reaction.

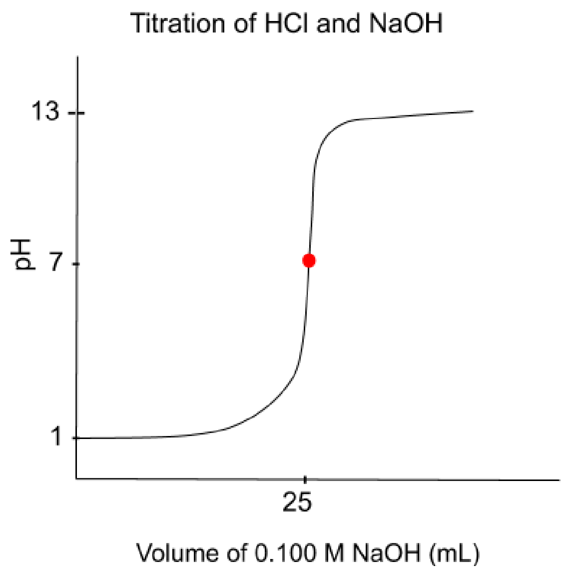
Endpoint: the point at which there is a visible change in the Erlenmeyer flask.

Acid-base titration: used to determine the concentration of an acid or base in a solution. The endpoint is usually indicated by a pH change, which can be detected using a pH meter or an indicator solution.

- Phenolphthalein: a common indicator for acid-base titrations. Turns the solution light pink at the equivalence point.
- If the analyte is a weak acid and the titrant is a strong base, the pH of the analyte is less than 7 before the titration begins. As the base is added, the solution's pH increases and changes color at the endpoint.

Redox titration: used to determine the concentration of an oxidizing or reducing agent in a solution. The endpoint is usually indicated by a color change, which can be detected using an indicator solution or an electronic meter.

Titration curve: a graphical representation of the relationship between the volume of titrant added and the corresponding pH of the analyte.



Linear region: the portion of the curve where the pH of the analyte is fairly constant as more titrant is being added to the solution.

Inflection point: the point on the curve where the slope changes. Indicates the equivalence point.

Endpoint: the point on the curve where the chemical reaction is complete. Indicated by a color change.

$5H_3AsO_3(aq) + 2MnO_4^-(aq) + 6H^+(aq) \rightarrow 5H_3AsO_4(aq) + 2Mn^{2+}(aq) + 3H_2O\ (l)$

The reaction between 5 $H_3AsO_3(aq)$ and $MnO_4^-(aq)$ in acidic solution is represented above. In a titration experiment, a 0.010 M solution of $KMnO_4(aq)$ is added from a buret to an acidified sample of $H_3AsO_3(aq)$ in a flask. The endpoint of the titration is determined by the appearance of a faint pink color in the mixture.

Initial buret reading	**2.57 mL**
Final buret reading	**42.57 mL**

Based on the data in the table above, how many moles of $H_3AsO_3(aq)$ were in the sample?

Determine the amount of $KMnO_4(aq)$ used in the titration. The *volume* of $KMnO_4$ used is the difference between the initial and final buret readings given in the table:

- 42.57 mL - 2.57 mL = 40.00 mL $KMnO_4$

Convert from volume to moles using the molarity of the $KMnO_4$ solution:

- $40.00\ mL\ KMnO_4 \times \frac{1\ L}{1000\ mL} \times \frac{0.010\ mol\ KMnO_4}{L\ KMnO_4} = 4.0 \times 10^{-4}\ mol\ KMnO_4$

Convert from moles of $KMnO_4$to moles of H_3AsO_3 using the mole ratio between the two substances:

- $4.0 \times 10^{-4}\ mol\ KMnO_4 \times \frac{5\ mol\ H_3AsO_3}{2\ mol\ KMnO_4} = 1.0 \times 10^{-3} mol\ H_3AsO_3$

So, there were 1.0×10^{-3} moles of H_3AsO_3 (aq) in the sample.

A student is given a solution containing 1.08 g of an unknown diprotic acid [an acid that yields two H^+ ions per acid molecule], H_2A. To determine the molar mass of the acid, the student titrated the sample using a 0.524 M solution of KOH(*aq*). The balanced equation for the reaction is shown below.

H_2A (aq) + 2KOH (aq) → $2H_2O$ (l) + K_2A (aq)

If 35.5 mL of KOH (aq) is required to consume the acid completely, what is the molar mass of the unknown acid? *Write your answer using three significant figures.*

Determine the amount of KOH used to consume the acid:

- $0.0355\ L\ soln \times \frac{0.524\ mol\ KOH}{L\ soln} = 1.86 \times 10^{-2} KOH$

Use the amount of KOH to find the amount of H_2A in the sample (2 moles of KOH are required for every 1 mole of H_2A):

- $1.86 \times 10^{-2}\ mol\ KOH \times \frac{1\ mol\ H_2A}{2\ mol\ KOH} = 9.30 \times 10^{-3}\ mol\ H_2A$

Use the amount of H_2A and its mass in the sample to calculate the molar mass of the acid:

- $\frac{1.08\ g\ H_2A}{9.30 \times 10^{-3}\ mol\ H_2A} = 116\ g/mol$

A 0.20 M solution of HBr (aq) is used to titrate 50 mL samples of four different metal hydroxide solutions. The volume of HBr (aq) required to reach the endpoint for each sample is shown in the table below.

Solution	Metal Hydroxide	Volume of HBr added (mL)
1	NaOH	35
2	KOH	60
3	$Sr(OH)_2$	20
4	$Ba(OH)_2$	75

Which solution contains the highest concentration of metal hydroxide?

Calculate the amount of metal hydroxide required to reach the endpoint of each titration.

1) $0.035\ L\ soln \times \frac{0.20\ mol\ HBr}{L\ soln} \times \frac{1\ mol\ NaOH}{1\ mol\ HBr} = 7.0 \times 10^{-3}\ mol\ NaOH$

2) $0.060\ L\ soln \times \frac{0.20\ mol\ HBr}{L\ soln} \times \frac{1\ mol\ KOH}{1\ mol\ HBr} = 1.2 \times 10^{-2}\ mol\ KOH$

3) $0.020\ L\ soln \times \frac{0.20\ mol\ HBr}{L\ soln} \times \frac{1\ mol\ Sr(OH)_2}{2\ mol\ HBr} = 2.0 \times 10^{-3}\ mol\ Sr(OH)_2$

4) $0.075\ L\ soln \times \frac{0.20\ mol\ HBr}{L\ soln} \times \frac{1\ mol\ Ba(OH)_2}{2\ mol\ HBr} = 7.5 \times 10^{-3}\ mol\ Ba(OH)_2$

Solutions 1 and 2 have a 1:1 mole ratio of HBr to metal hydroxide, while 3 and 4 have a 2:1 ratio. This is because $Sr(OH)_2$ and $Ba(OH)_2$ both produce *two* equivalents of OH^- per mole of base.

The solution containing the highest concentration of metal hydroxide is solution 2.

Unit 5: Kinetics

5.1: Reaction Rates

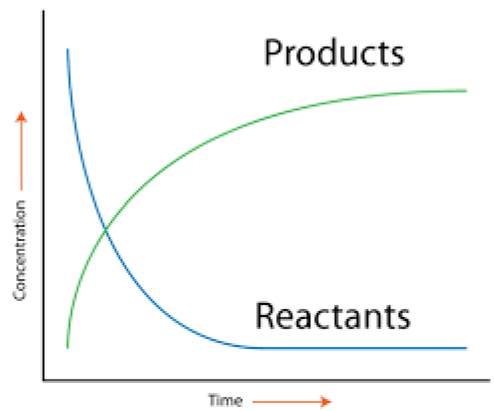
[Reaction Rates - AP Chem Study Guide 2024 | Fiveable]

Kinetics: the study of the rate of a reaction.

Rate of reaction: how quickly a reaction produces products. Measured by observing concentrations. (units: mol/Ls)

- rate = $-\frac{\Delta[Reactant]}{t}$ OR rate = $\frac{\Delta[Product]}{t}$

Average rate of reaction: divide the change in concentration by the time interval over which the change in concentration occurred.



Instantaneous rate of a reaction: the rate of the reaction at a specific point in time. It is calculated by taking the limit of the average rate as the time interval approaches zero.

- Rate = -d[R]/dt

Collision theory (the reacting particles must collide to react):

1) Concentration: increasing the concentration of reactants increases the rate of the reaction. There are more reactant molecules present.
2) Temperature: increasing the temperature of reactants generally increases the rate of a reaction. An increase in temperature means that the reactant molecules have more kinetic energy.
3) Surface area: increasing the surface area of a reactant can increase the rate of a reaction. A larger surface area means more surface available for reactant molecules to collide with.
4) Presence of a catalyst: the catalyst provides an alternative pathway for the reaction to occur, which can lower the activation energy needed for the reaction to take place.
5) Pressure: in reactions involving gases, increasing the pressure can increase the rate of the reaction. There are more gas molecules present in a given volume.

At a certain time during a reaction, the rate of disappearance of substance X was 3.0 x 10^{-3} M s^{-1}, and the rate of disappearance of substance Y was 1.5 x 10^{-3} M s^{-1}. At the same time, the rate of appearance of substance Z was 3.0 x 10^{-3} M s^{-1}.

Which of the following represents the stoichiometry of the reaction?

The rates of disappearance or appearance of the substances in a reaction depend on their stoichiometric coefficients in the balanced equation.

X disappears twice as fast as substance Y does, meaning that two atoms/molecules of substance X are consumed for each atom/molecule of substance Y consumed.

Substance X disappears at the same rate that substance Z appears, meaning that one atom/molecule of substance X is consumed for each atom of substance Z produced.

2X + Y → 2Z.

$2NO_2$ (g) → 2NO (g) + O_2 (g)

- $rate = -\frac{1}{2}\frac{\Delta[NO_2]}{\Delta t} = -\frac{1}{2}\frac{(0.532\ M - 0.654\ M)}{120.0\ s - 60.0\ s} = \frac{1.02\times10^{-3} M}{s}$

$2H_2O_2$ (aq) → $2H_2O$ (l) + O_2 (g)

In the presence of a catalyst, H_2O_2(*aq*) decomposes according to the equation above. A sample of H_2O_2(*aq*) is monitored as it decomposes, and the concentration of H_2O_2 as a function of time is recorded.

Time (s)	$[H_2O_2]$
0	1.000
50.0	0.766
100.0	0.587
150.0	0.450

Calculate the average rate of the reaction between 100.0 and 150.0 seconds. ***Write your answer in scientific notation using three significant figures.***

$rate = -\frac{1}{2}\frac{\Delta[H_2O_2]}{\Delta t} = -\frac{1}{2}\frac{(0.450\ M - 0.587\ M)}{(150.0\ s - 100.0\ s)} = 1.37 \times 10^{-3}\ M/s$

5.2: Introduction to Rate Law

https://library.fiveable.me/ap-chem/unit-5/rate-law-intro/study-guide/oq5mJS35IadrTLHLjdqZ

Rate law: an equation that describes the relationship between the rate of a chemical reaction and the concentration of the reactants. [$R = k[A]^n[B]^m$...]

- R: the rate of the reaction (see 5.1!)
- k: the rate constant (temperature-dependent!)
- n and m: the reaction orders for each reactant (A, B, etc).

Reaction order: how the rate of the reaction changes as the concentration of each reactant changes.

- Ex: A + B → C is $R = k[A]^2[B]^1$
- If the concentration of A doubles, the rate will quadruple. If the concentration of B is doubled, the rate will double. (Assuming the other reactant(s) remain constant.)

Overall reaction order: the sum of the orders for each reactant.

$2HgCl_2\,(aq) + C_2O_4^{2-}\,(aq) \rightarrow 2Cl^-\,(aq) + 2CO_2\,(g) + Hg_2Cl_2\,(aq)$

A rate study of the reaction represented above yields the following data:

Trial	Initial $[HgCl_2]$ (M)	Initial $[C_2O_4^{2-}]$ (M)	Initial reaction rate (M s^{-1})
1	0.020	0.010	3.0×10^{-5}
2	0.020	0.020	1.2×10^{-4}
3	0.010	0.020	6.0×10^{-5}

Based on the data, what is the rate law for the reaction?

M (the order of the reaction for $HgCl_2$) should equal 1 as the results from trials 2 and 3 show that when $[HgCl_2]$ is halved (with $[C_2O_4^{2-}]$ constant), the rate of the reaction is also halved.

N (the order of the reaction with respect to $C_2O_4^{2-}$) should equal 2 as the results from trials 1 and 2 show that when oxalate is doubled (with $[HgCl_2]$ constant, the rate of the reaction is multiplied by four ($=2^2$).

So, the rate law for the reaction is rate = $k[HgCl_2][C_2O_4^{2-}]^2$.

$2X (g) + 2Y (g) \rightarrow Q (g) + 2R (g)$

The reaction represented above is found to be second order with respect to X and first order with respect to Y.

What happens to the rate of the reaction when [X] is halved and [Y] is doubled?

rate = $k[X]^2[Y]$

Since [X] is raised to the second power in the rate law, halving [X] *decreases* the reaction rate by a factor of 4.

- $[\frac{1}{2}X]^2 = \frac{1}{4}[X]^2$

Since [Y] is raised to the first power, doubling [Y] *increases* the reaction rate by a factor of 2.

Combined, these changes result in the reaction rate decreasing by a factor of 2 overall.

Zeroth-order reaction: the reaction rate does not depend on the reactant concentration. A linear change in concentration with time is a clear indication of a zeroth-order reaction.

5.3: Concentration Changes Over Time

AP Chem – 5.3 Concentration Changes Over Time | Fiveable

First-Order Reactions (Chemical Kinetics) - BYJU'S

Integrated rate law: the relationship between concentrations of reactants and time.

- For zeroth-order reactions: $[A]_t - [A]_0 = -kt$
- For first-order reactions (including radioactive decay): $\ln[A]_t - \ln[A]_0 = -kt$
- For second-order reactions: $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$

All three integrated rate laws are in the form y - b = mx (from y = mx + b) where y is the current concentration, b is the concentration before any time has passed, m is the rate constant (k), and x is time (t).

Half-life: the time taken for the radioactivity of a specified isotope to fall to half its original value. [$t_{1/2} = 0.693/k$]

The radioisotope ^{32}P decays to ^{32}S with a half-life of 14.3 days. If a sample initially contains 18.8 mg of ^{32}P, what mass of ^{32}P remains after 21.0 days? ***Write your answer using three significant figures.***

Use the half-life equation to determine the value of the rate constant (k):

- $t_{1/2} = 0.693/k = 0.693/14.3\ d = 0.0485\ d^{-1}$
- d = days (source)

Use the first-order integrated law to find the final mass of ^{32}P:

- $\ln(N_t) - \ln(N_0) = -kt \rightarrow \ln(N_t/N_0) = -kt$ [quotient property of logarithms]
- $N_t/N_0 = e^{-kt} \rightarrow N_t = N_0 e^{-kt} = 18.8\ mg[e^{-(0.0485\ d^{-1})(21.0\ d)}] = 6.79\ mg^*$
- *Since the mass of a radioisotope sample (m) is proportional to the number of radioactive nuclei in the sample (N), we can substitute the ratio m_t/m_0 for the ratio N_t/N_0 in the integrated rate equation. This allows us to work with initial and final masses instead of initial and final numbers of nuclei in our calculations.

HCO_2H (g) → CO_2 (g) + H_2 (g)

The first-order decomposition of HCO_2H (g) is represented by the equation above. At a certain temperature, the partial pressure of HCO_2H (g) in a sealed vessel falls from 300. Torr to 75.0 Torr over 240. seconds. What is the half-life of the decomposition reaction?

The half-life of a reaction is the time required for a reactant to reach one-half its initial concentration or pressure. For a first-order reaction, the half-life is constant over time.

The partial pressure of HCO_2H (g) fell to ¼ its initial value over 240. Seconds. This means that two half-lives must have occurred during that time.

So, the half-life of the decomposition reaction is 120. seconds.

5.4: Reaction Mechanisms

[15.2: The Equilibrium Constant (K) - Chemistry LibreTexts]

Elementary reaction: a reaction that occurs in a single step with only one transition state. Represents an actual interaction between reactant particles, so the rate law can be derived from its balanced equation.

- $2NO_2\,(g) + O_3\,(g) \rightarrow N_2O_5\,(g) + O_2\,(g)$ rate = $k[NO_2]^2[O_3]$

Reaction mechanism: the sequence of elementary steps by which a chemical reaction occurs.

Multistep (complex) reaction: a reaction that occurs in two or more elementary steps.

- The equations for the elementary steps in the mechanism must add up to the overall equation for the reaction.
- The mechanism must be consistent with the experimental rate law.

Reaction intermediate: a species that is formed in one step and consumed in a subsequent step.

Rate-determining step: the slowest step in the mechanism [consumes the most energy!]. No reaction can occur faster than its slowest step, so this limits the overall rate of the reaction.

Molecularity of an elementary step: the number of reactant particles that participate in the step.

- Unimolecular: if one reactant particle is involved.
- Bimolecular: if two reactant particles are involved.

Chemical equilibrium: the forward reaction rate equals the reverse reaction rate. The net composition of the system no longer changes with time.

Pre-equilibrium approximation: a method used to derive the rate law for a mechanism with a fast initial step.

1. Write the rate law based on the slow (rate-determining) step.
2. Eliminate any intermediates from the rate law by using the fast initial step to solve for the intermediate concentration(s) in terms of reactant and/or product concentrations.

A proposed mechanism for the decomposition of NO_2Cl (g) is represented below.

Step 1: NO_2Cl (g) ⇋ NO_2 (g) + Cl (g) (fast equilibrium)

Step 2: NO_2Cl (g) + Cl (g) → NO_2 (g) + Cl_2 (g) (slow)

Overall: $2NO_2Cl$ (g) → $2NO_2$ (g) + Cl_2 (g)

What rate law for the overall reaction is consistent with the proposed mechanism?

Write the rate law for the slow step in the mechanism. In this case, the slow step is step 2, so the rate law (derived from the stoichiometry of the balanced equation) is:

- rate = $k_2[NO_2Cl][Cl]$

(k_2 indicates that this k refers to the rate constant of the second step)

Solve for the concentration of the intermediate Cl in terms of reactant and/or product concentrations. To do so, assume that the fast initial step in the mechanism is at equilibrium:

- rate = $k_1[NO_2Cl] = k_{-1}[NO_2][Cl]$

Solving for [Cl], we get:

- $[Cl] = \frac{k_1[NO_2Cl]}{k_{-1}[NO_2]}$

Substitute the expression for [Cl] into the rate law for the second step:

- rate = $k_2[NO_2Cl][Cl]$
- $= k_2[NO_2Cl]\frac{k_1[NO_2Cl]}{k_{-1}[NO_2]}$
- $= k_2\frac{k_1[NO_2Cl]^2}{k_{-1}[NO_2]}$
- $= k\frac{[NO_2Cl]^2}{[NO_2]}$

5.5: Activation Energy and Reaction Rate

AP Chem Unit 5.5 Collision Model | AP Chemistry | Fiveable

Activation energy (article) | Khan Academy

Arrhenius Equation - Expression, Explanation, Graph, Solved Exercises

Exergonic: accompanied by the release of energy.

Endergonic: accompanied by or requiring the absorption of energy.

Activation energy (E_a): the energy needed to begin a reaction.

Overall energy change (ΔE): the difference between final and initial energy.

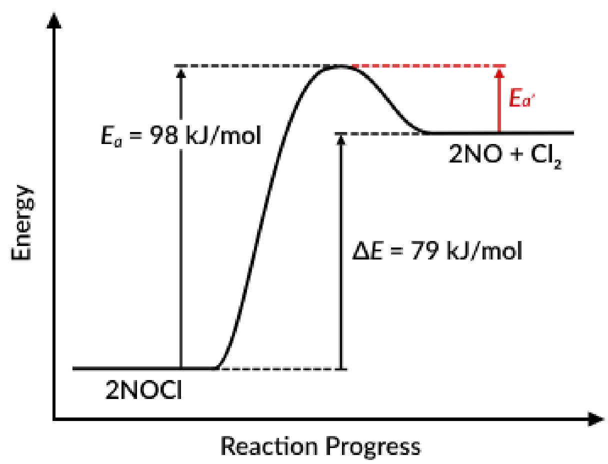
Activation energy for the reverse step ($E_{a'}$): the difference between the activation energy and the overall energy.

$2NOCl\ (g) \rightarrow 2NO\ (g) + Cl_2\ (g)$

The reaction represented above occurs in a single elementary step. The activation energy for the step is 98 kJ/mol, and the overall energy change is 79kJ/mol.

What is the activation energy for the reverse step?

98 kJ/mol - 79 kJ/mol = 19 kJ/mol



Collision model: models the collisions of molecules. As they collide at fixed average speeds determined by temperature, they bounce off one another, preserving their kinetic energy and momentum.

- Ineffective collision: occurs when there isn't enough force, molecules are moving too slowly, and/or the molecules aren't aligned right. This is why some reactions at different temperatures move at different rates.
- Effective collision: occurs only when particles have sufficient energy ($E \geq E_a$) and arrange themselves in the proper orientation.

In a two-step mechanism, the rate-determining step is the elementary step with the *greater* activation energy.

Arrhenius equation: relates absolute temperature [K] and the rate constant.

- $k = Ae^{-E_a / RT}$

k	A	e
Rate constant of the reaction	The pre-exponential factor	Base of the natural number
E_a	R	T
Activation energy	Universal gas constant	Absolute temperature (in K)

- If the activation energy is expressed in terms of energy per reactant molecule, the universal gas constant must be replaced with the Boltzmann constant (k_B).

5.6: Catalysis

Catalyst: a substance that increases the rate of a reaction by decreasing the necessary activation energy without being consumed in the reaction.

- Enzymatic catalyst (enzymes): a biological catalyst that speeds up biochemical reactions in living organisms.
- Homogeneous catalyst: a catalyst that exists in the same phase as the reactants.
- Heterogeneous catalyst: a catalyst that exists in a different phase than the reactants.

“A phase of matter is characterized by having relatively uniform chemical and physical properties. Phases are different from states of matter.

The states of matter (e.g., liquid, solid, gas) are phases, but matter can exist in different phases yet remain in the same state of matter. For example, liquid mixtures can exist in multiple phases, such as an oil phase and an aqueous phase.” - thoughtco.com

Unit 6: Thermodynamics

6.1: Endothermic and Exothermic Processes

AP Chem Unit 6.1 Endothermic & Exothermic Processes | AP Chemistry | Fiveable

The laws of thermodynamics (article) | Khan Academy

System: a specific part of the universe that is of interest to us and would typically involve substances used in experiments

- Open system: can exchange both energy and matter with its surroundings.
- Closed system: can exchange only energy with its surroundings, not matter.
- Isolated system: cannot exchange either matter or energy with its surroundings.

Surroundings: everything not in the defined system.

Universe: the system and surroundings.

Entropy: the degree of randomness or disorder in a system.

Enthalpy (H): a state function that describes the total heat content of a system.

- ΔH (Delta H): the change in enthalpy of a system or the difference between the final and initial enthalpies of a system. **A measure of the heat absorbed or released in a chemical reaction or a physical process such as a change of phase or temperature.**
- If ΔH is positive (energy added), the process absorbs heat from the surroundings and is said to be endothermic.
- If ΔH is negative (energy released), the process releases heat to the surroundings and is said to be exothermic.

The first law of thermodynamics: $\Delta E = Q - W$

- ΔE = the internal energy of a system.
- Q = the heat transferred (equal to the change in enthalpy (ΔH) at constant pressure).
- W = the work done by the system.

6.2: Heat Transfer and Thermal Equilibrium

[AP Chem Unit 6.2 Energy Diagrams of Reactions | AP Chemistry | Fiveable]

[AP Chem Unit 6.3 Kinetic Energy, Heat Transfer & Thermal Equilibrium | AP Chemistry | Fiveable]

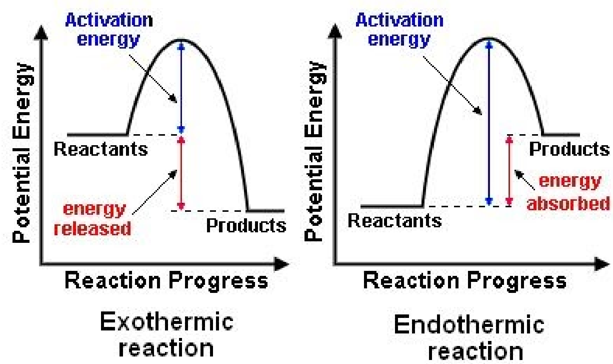
Thermodynamic equilibrium (thermal equilibrium): kinetic energy (in the form of heat) is transferred from warm to cold until they are equal. The molecules are moving at the same speed since the temperatures are the same.

Energy diagrams (potential energy diagrams): show the potential energy of the reactants and products, as well as the activation energy required for the reaction to occur.

- PEreactants: the potential energy of the reactants at the beginning of the reaction before any energy is added or released.
- PEproducts: the potential energy of the products at the end of the reaction after all the energy has been added or released.
- Activation energy: the minimum amount of energy required to initiate a chemical reaction; the amount of energy required to overcome the initial barriers that prevent the reactants from reacting.
- Activated complex: a hypothetical intermediate state that a molecule must pass through before it can react. It is the highest point on the energy diagram, representing the most unstable state of the reactants before they transform into products. The reaction occurs when the energy supplied to the reactants exceeds the activation energy and the system reaches the activated complex.

The energy diagram for an endothermic process: upward slope (energy is being absorbed by the system as the reaction or phase change proceeds).

The energy diagram for an exothermic process: downward slope (energy is being released by the system as the reaction or phase change proceeds).



6.3: Heat Capacity and Calorimetry

AP Chem Unit 6.4 Heat Capacity & Calorimetry | AP Chemistry | Fiveable

Calorimetry: the study of heat flow and heat exchange between a system and its surroundings. Can be used to calculate ΔH by measuring changes in temperature, which represent heat being lost or gained.

- Calorimeters: tools used to measure the heat flow in a chemical reaction or physical process.

Specific heat: the amount of heat required to raise the temperature of a gram of a substance by 1° C.

- Heat capacity: the amount of heat required to raise the temperature of an object by 1° C.

$q = mc\Delta T$: quantifies the magnitude of heat transferred between the system and its surroundings in a calorimeter.

- q: the heat in Joules.
- m: the mass in grams or kilograms.
- c: the specific heat of the substance.
- ΔT: the change in temperature in Kelvin OR Celsius (difference is the same).

A sample of peanut is combusted directly below a can containing 250. g of water initially at 24.0° C. The reaction releases 4.2 kJ of heat energy, all of which is transferred to the water.

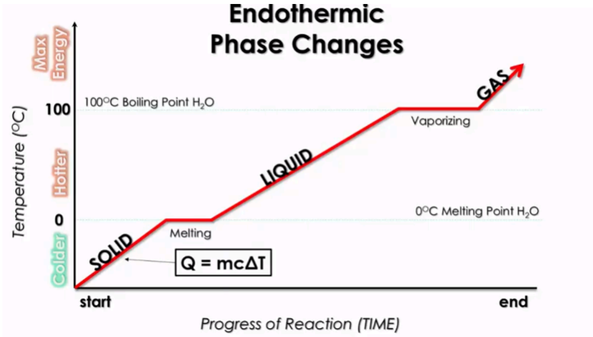
What is the final temperature of the water? (The specific heat of water is 4.2 J/(g x °C).)

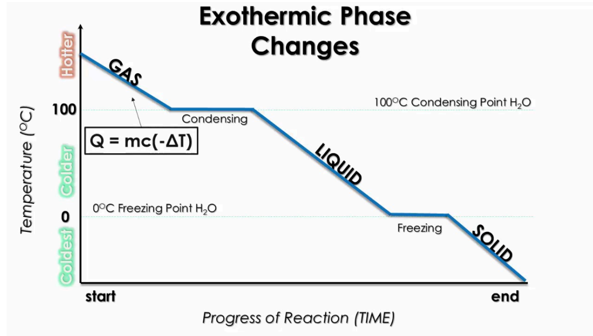
4200 J = 250. g x 4.2 (J/g x °C) x ΔT → ΔT = 4.0 °C

T_{final} = 4.0 °C + 24.0 °C = 28.0 °C

6.4: Energy of Phase Changes

[AP Chem Unit 6.5 Phase Changes & Energy | AP Chemistry | Fiveable]





Heating curves show temperature changes over time as a substance heats up; cooling curves are the inverse and show exothermic processes.

Plateaus: occur when energy is being transferred at the same rate it is being used for the phase change. Thus, the temperature stays constant.

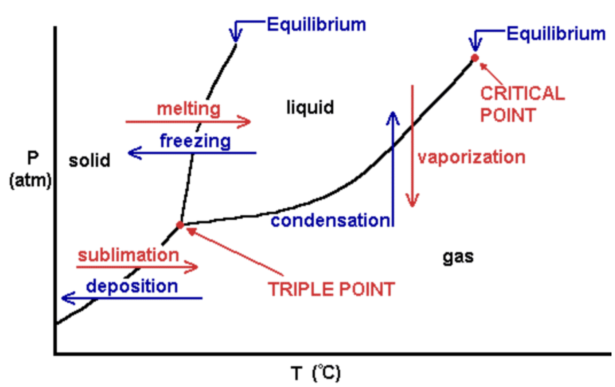
- Heats of fusion (H_f): the energy needed to liquify a solid at its melting point without changing its temperature (334 J/g for H_2O).
- Heats of vaporization (H_v): the energy needed to turn a liquid to gas at its boiling point w/out changing its temperature (2260 J/g for H_2O).
- Heats of condensation: the negative of heats of vaporization.
- Heats of freezing: the negative of heats of fusion.

When calculating energy during phase changes, use q=mcΔT at the slopes (since there is a change in temperature) and H_f(m) or H_v(m) on the plateaus. [m = mass in grams]

The phase diagram shows the relationship between temperature and pressure in determining what state of matter an object is in.

Triple point: a weird point where you are in all three states of matter.

Critical point: a point at which past this point you can no longer have a liquid, you either have a supercritical fluid or a gas.



6.5: Introduction to Enthalpy of Reaction

Units in Thermochemical Calculations – AP Central

mol_{rxn} (mole of reaction): reaction unit. For a chemical reaction, the change in moles for each species is tied to mol_{rxn} by the stoichiometric coefficient.

- Given the equation $Mg + 2HCl \rightarrow MgCl_2 + H_2$, 0.100 moles of HCl can react with excess Mg to produce $0.100\ mol\ HCl \times \frac{1\ mol_{rxn}}{2\ mol\ HCl} = 0.05\ mol_{rxn}$

Methane, CH_4 (*g*) (molar mass 16.0 g/mol), is a gas commonly used to fuel Bunsen burners. The combustion of CH_4 (*g*) is represented by the equation below.

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ $\Delta H°_{rxn}$* = -891 kJ/mol_{rxn}

Calculate the mass of CH_4 (*g*) that would need to be combusted to heat 500.0 g of water from 22.0 °C to 100.0 °C. (Assume that all of the heat released by the reaction is absorbed by the water and that the specific heat of water is 4.18 J/(g x °C).)

*****The small circle on the $\Delta H°_{rxn}$ signifies that the given enthalpy is under STP (1 atm & 25 °C).

- Calculate the amount of heat required to warm the water:

  q_{water} = 500.0 g x (4.18 J/(g x °C)) x (100.0 °C - 22.0 °C) = 163,000 J = 163 KJ

- Find the mass of CH_4 (*g*) that would produce this amount of heat when combusted:
  - Since all of the heat released by the reaction is assumed to be absorbed by the water:

    $q_{rxn} = -q_{water}$ = -163 kJ

  - Use the $\Delta H°_{rxn}$ and the molar mass of methane to convert from kilojoules to grams:

    $$-163\ kJ \times \frac{1\ mol_{rxn}}{-891\ kJ} \times \frac{1\ mol\ CH_4}{1\ mol_{rxn}} \times \frac{16.0\ g\ CH_4}{1\ mol\ CH_4} = 2.93\ g\ CH_4$$

When a 50.0 mL sample of 0.20 M $MgCl_2$ (aq) and a 50.0 mL sample of 0.20 M Na_2CO_3 (aq) are mixed in a calorimeter, a reaction occurs according to the equation below.

$MgCl_2$ (aq) + Na_2CO_3 (aq) → $MgCO_3$ (s) + 2NaCl (aq)

If the temperature of the mixture decreases from 22.0 °C to 20.8 °C, what is the experimental value of ΔH^o_{rxn} for the reaction? (Assume that the mixture has a total mass of 100. G and a specific heat of 4.2 J/(g x °C).)

- Calculate the amount of heat lost from the solution during the experiment (q_{soln}):

  q_{soln} = mcΔT = (100. g)(4.2 J/(g x °C))(20.8 °C - 22.0 °C) = -500 J = -0.50 kJ

- Since the reaction takes place in a calorimeter, all the heat lost from the solution is assumed to be absorbed by the reaction:

  $q_{rxn} = -q_{soln}$ = 0.50 kJ

- Since both reactants have a stoichiometric coefficient of 1, dividing q_{rxn} by the number of moles of either reactant will give us ΔH^o_{rxn} per mole of reaction.

$$\Delta H^{o}_{rxn} = \frac{q_{rxn}}{nMgCl_2} = \frac{0.50\ kJ}{0.010\ mol\ MgCl_2} = 50.\ kJ/mol_{rxn}$$

6.6: Hess's Law

[Hess's Law - Chemistry LibreTexts]

Hess's law: the heat of any reaction ΔH°_f for a specific reaction is equal to the sum of the heats of reaction for any set of reactions which in sum are equivalent to the overall reaction.

Reaction	Equation	ΔH° (kJ/mol_{rxn})
W	$N_2\,(g) + 3H_2\,(g) \rightarrow 2NH_3\,(g)$	-92
X	$2H_2\,(g) + O_2\,(g) \rightarrow 2H_2O\,(l)$	-572
Y	$N_2\,(g) + 2O_2\,(g) \rightarrow 2NO_2\,(g)$	66
Z	$2NH_3\,(g) + 4H_2O\,(l) \rightarrow 2NO_2\,(g) + 7H_2\,(g)$	?

Using the information above, determine the value of ΔH°_{rxn} for reaction Z.

- The equation for reaction W has 2 mol of NH_3 (g) as a product, whereas the equation for reaction Z has 2 mol of NH_3 (g) as a reactant. So, reverse the equation for reaction W and change the sign of the ΔH° value:

  $2NH_3\,(g) \rightarrow N_2\,(g) + 3H_2\,(g) \qquad \Delta H^\circ = 92\ kJ/mol_{rxn}$

- The equation for reaction X has 2 mol of H_2O (l) as a product, whereas the equation for reaction Z has 4 mol of H_2O (l) as a reactant. So, reverse the equation for reaction X, change the sign of the ΔH° value, and multiply the equation and the ΔH° value by 2:

  $2[2H_2O\,(l) \rightarrow 2H_2\,(g) + O_2\,(g)] \qquad \Delta H^\circ = 2(572\ kJ/mol_{rxn})$

- The equation for reaction Y has 2 mol of NO_2 (g) as a product, which is consistent with the equation for reaction Z. So, leave the equation for reaction Y unchanged.

- Add the three equations and their enthalpy changes:

$$
\begin{array}{rr|l}
 & 2NH_3\,(g) \rightarrow \cancel{N_2\,(g)} + 3H_2\,(g) & \Delta H^\circ = 92\ kJ/mol_{rxn} \\
 & 4H_2O\,(l) \rightarrow 4H_2(g) + \cancel{2O_2\,(g)} & \Delta H^\circ = 1144\ kJ/mol_{rxn} \\
+ & \cancel{N_2\,(g)} + \cancel{2O_2\,(g)} \rightarrow 2NO_2\,(g) & \Delta H^\circ = 66\ kJ/mol_{rxn} \\
\hline
 & 2NH_3\,(g) + 4H_2O\,(l) \rightarrow 2NO_2\,(g) + 7H_2\,(g) & \Delta H^\circ_{rxn} = 1302\ kJ/mol_{rxn}
\end{array}
$$

6.7: Enthalpy of Formation

[8.4: Standard States and Enthalpies of Formation - Chemistry LibreTexts]

I. Standard state: the most stable form of a substance at a pressure of one bar at any particular temperature.
   - For water at -10 °C, the standard state is ice at the pressure of one bar. At +10 °C, it is liquid water at a pressure of one bar.
   - Standard state of a gas: the ideal gas standard state at that temperature.

II. Standard enthalpy of formation (standard heat of formation): the enthalpy change for a reaction in which **the product is one mole of the substance** and **the reactants are the compound's constituent elements in their standard states**.
   - For water at -10 °C, this reaction is:
     H_2 (g, -10 °C, 1 bar) + ½O_2 (g, -10 °C, 1 bar) → H_2O (s, -10 °C, 1 bar).
   - For water at +10 °C, it is:
     H_2 (g, +10 °C, 1 bar) + ½O_2 (g, +10 °C, 1 bar) → H_2O (liq, + 10 °C, 1 bar).
   - For water at +110 °C, it is:
     H_2 (g, +110 °C, 1 bar) + ½O_2 (g, +110 °C, 1 bar) → H_2O (g, +110 °C, 1 bar).

III. The standard enthalpy of formation is given the symbol ΔH_f^o.
   - The superscript degree sign indicates that the reactants and products are all in their standard states.
   - The subscript, f, indicates that the enthalpy change is for the formation of the indicated compound from its elements.

IV. The standard enthalpy change of a reaction is related to the standard enthalpies of formation of the reactants and products by the following equation: $\Delta H^o_{rxn} = \Sigma \Delta H^o_f(products) - \Sigma \Delta H^o_f(reactants)$.

V. Standard enthalpy of formation for any element at any particular temperature: zero.
   - By definition, the standard enthalpy of formation is the enthalpy change where the reactants are the **constituent elements** of the product. Thus, it takes no energy to produce an element from itself (O_2 → O_2, H_2 → H_2, etc.). [mildly questionable source]

The standard enthalpy change for the reaction represented below is ΔH_1. The standard enthalpies of formation for B_2O_3 (s) and H_2O (g) are ΔH_2 and ΔH_3, respectively.

B_2H_6 (g) + $3O_2$ (g) → B_2O_3 (s) + $3H_2O$

What expression is equivalent to the standard enthalpy of formation for B_2H_6 (g)?

- Write the equation for the standard enthalpy change for the reaction between B_2H_6 (g) and O_2:

$$\Delta H^o_{rxn} = [\Delta H^o_{f,\,B_2O_3\,(s)} + 3(\Delta H^o_{f,\,H_2O\,(g)})] - [\Delta H^o_{f,\,B_2H_6\,(g)} + 3(\Delta H^o_{f,\,O_2\,(g)})]$$

- Plug in the given enthalpy changes and solve for $\Delta H^o_{f,\,B_2H_6\,(g)}$. Since the enthalpy of formation for an element in its standard state is 0, we know that $\Delta H^o_{f,\,O_2\,(g)} = 0$.

$$\Delta H^o_{f,\,B_2H_6\,(g)} = \Delta H_2 + 3\Delta H_3 - 3(0) - \Delta H_1 = \Delta H_2 + 3\Delta H_3 - \Delta H_1$$

6.8: Bond Enthalpies

The enthalpy change of a reaction is related to the enthalpies of the bonds broken and formed in the reaction by the following equation:

$$\Delta H^{o}_{rxn} = \Sigma(enthalpies\ of\ bonds\ broken) - \Sigma(enthalpies\ of\ bonds\ formed)$$

Bonds broken = on the reactants side

Bonds formed = on the products side

Unit 7: Equilibrium

7.1: Introduction to Equilibrium

Reversible reaction: a reaction that can proceed in both the forward and backward directions.

(Dynamic) equilibrium: the forward and reverse processes occur at the same rate, resulting in no observable change in the system (the concentrations/partial pressures of all species remain constant).

If the rate of the forward reaction is greater than the rate of the reverse reaction, there is a net conversion of reactants to products.

If the rate of the forward reaction is less than the rate of the reverse reaction, there is a net conversion of products to reactants.

If the two rates are equal, the reaction is at equilibrium.

7.2: Equilibrium Constant and Reaction Quotient

The equilibrium constant K (article) | Khan Academy

$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$, where aA + bB ⇌ cC + dD

- K_c (also known as K or K_{eq}): the equilibrium constant using molar concentrations
- Remember, solids and liquids have a value of 1!

$K_p = \frac{[P_C]^c[P_D]^d}{[P_A]^a[P_B]^b}$

- K_p: equilibrium constant using gas pressures

For reactions that are not at equilibrium, we can write a similar expression called the reaction quotient (Q), which is equal to K_c when the reaction is at equilibrium.

7.3: Calculating the Equilibrium Constant

X (g) + Y (g) ⇋ 2Z (g) + Q (g)

A 6.00 mol sample of X (g) and a 6.00 mol sample of Y (g) are combined in a rigid, 1.00 L container at a particular temperature, and the reaction represented above occurs. When the mixture reaches equilibrium, 5.00 mol of Q (g) is present in the container.

What is the value of the equilibrium constant, K_c, for the reaction at this temperature?

Let's use the balanced equation to write out the equilibrium constant expression:

- $K_c = \frac{[Z]^2[Q]}{[X][Y]}$

5.00 mol of Q (g) is present in the 1.0 L container at equilibrium, so [Q] = 5.00 M.

Since 1 mol of both X (g) and Y (g) is consumed for every 1 mol of Q (g) formed, [X] = [Y] = 1.00 M.

Because 2 mol of Z (g) is formed for every 1 mol of Q (g) formed, [Z] = 5.00 x 2 = 10.0 M.

Plug these values into our expression and solve for K_c:

- $K_c = \frac{(10.0)^2(5.00)}{(1.00)(1.00)} = 500.$

7.4: Magnitudes and Properties of the Equilibrium Constant

A large K value (greater than 1) indicates that there are more products than reactants at equilibrium.

A small K value (less than 1) indicates that there are more reactants than products at equilibrium.

Properties of the equilibrium constant:

- If the equation is reversed, K is inverted.
- If the equation is multiplied by a factor of n, K is raised to the n^{th} power.
- If multiple equations are added together, K for the overall equation is the product of the K values for the equations that were summed.

7.5: Calculating Equilibrium Concentrations

I_2 (g) + Br_2 (g) ⇋ 2IBr (g) K_p 73.3 at 475 K

A rigid vessel initially contains I_2 (g) and Br_2 (g), each at a partial pressure of 2.00 atm at 475 K. The system comes to equilibrium according to the equation above.

Calculate the partial pressure of IBr (g) in the vessel at equilibrium at 475 K.

To solve this problem, we first need to use an ICE (Initial, Change, and Equilibrium) table to derive expressions for the equilibrium partial pressures of I_2 (g), Br_2 (g), and IBr (g) in terms of x.

Then, we can plug these expressions into the equilibrium constant expression to find the value of x, which we can in turn use to calculate P_{IBr}.

Let's start by setting up our ICE table:

	I_2 (g)	+	Br_2 (g)	⇌	2IBr (g)
Initial (atm)	2.00		2.00		0
Change (atm)	-x		-x		+2x
Equil. (atm)	2.00 - x		2.00 - x		2x

Next, let's find the value of x. To do so, we'll substitute the equilibrium concentrations from our ICE table into the expression for K_p:

- $K_p = 73.3 = \frac{(P_{IBR})^2}{(P_{I2})(P_{Br2})} = \frac{(2x)^2}{(2.00 - x)^2}$
- $\sqrt{73.3} = \frac{2x}{2.00 - x}$
- x = 1.62

P_{IBR} = 2x = 3.24 atm

*Notice: you <u>cannot</u> drop the x here because K_p is large!

7.6: Using the Reactant Quotient

$4NO_2$ (g) + O_2 (g) ⇋ $2N_2O_5$ (g)

At 325 K, 2.00 mol of each of the three gases in the reaction represented above are placed in an evacuated, rigid 1.00 L container. The value of K_c for the reaction at 3235 K is 4.57 x 10^{-3}.

At equilibrium, which of the gases will have the highest concentration?

To identify which gas will have the highest concentration at equilibrium, we first need to determine how the system will proceed to reach equilibrium. We can do so by calculating the reaction quotient, Q_c, and then comparing it with the value of K_c given in the problem.

$$Q_c = \frac{[N_2O_5]^2}{[NO_2]^4[O_2]} = \frac{[2.00]^2}{[2.00]^4[2.00]} = 0.125$$

Since $Q_c > K_c$, the system will increase [NO_2] and [O_2] and decrease [N_2O_5] until $Q_c = K_c$. Given that all three gases have the same initial concentration and that NO_2 (g) is produced four times as fast as O_2 (g), this means that [NO_2] will be higher than both [N_2O_5] and [O_2] at equilibrium.

7.7: Le Châtelier's Principle

If a reactant is added to the reaction at equilibrium, the reaction will shift right (towards the products) to reduce the amount of reactant and increase the amount of product (and vice versa if product is added).

Adding an inert gas (a gas not involved in the reaction) has no effect on equilibrium.

Reducing the volume of the container will shift the system to where there are fewer molecules of gas.

Increasing the volume of the container will shift the system to where there are more numbers of gaseous molecules.

Effect of change in temperature:

- If the reaction is endothermic, heat is a reactant in the reaction.
- If the reaction is exothermic, heat is a product.

$CO\ (g) + H_2O\ (g) \leftrightharpoons CO_2\ (g) + H_2\ (g)$

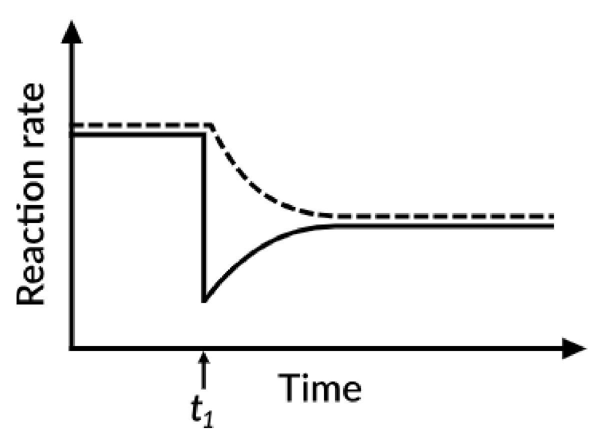
After the system above reaches equilibrium, a small amount of H_2O (g) is removed from the reaction vessel at a time t_1 while the temperature is held constant.

Which of the following graphs best shows the rates of the forward and reverse reactions as a function of time?

Removing H_2O will decrease the rate of the forward reaction sharply. Then, as H_2O (g) and CO (g) begin to build up in the system due to the reverse reaction, the forward rate will increase, eventually reaching a new constant value.

Initially, the reverse reaction rate will not be affected by the removal of H_2O (g). However, since CO_2 (g) + H_2 (g) are no longer being produced at the same rate by the forward reaction, the rate of the reverse reaction will begin to decrease.

This will occur until the reverse rate reaches the same constant value as the forward rate, at which point a new equilibrium position will be established.



7.8: Solubility Equilibria

K_{sp} (solubility constant): an equilibrium constant that reflects the extent to which an ionic compound dissolves in water.

If $Q < K_{sp}$, the newly mixed solution is undersaturated and no precipitate will form. If $Q > K_{sp}$, the solution is oversaturated and a precipitate will form until $Q = K_{sp}$.

Common-ion effect: the solubility of an ionic compound is decreased by the presence of a common ion (an ion that is also present in the compound).

- The common ion is a product, so if there is product already present in the solution, there will be less dissolution (forward reaction).

For ionic compounds containing basic anions, solubility increases as the pH of the solution is decreased.

For ionic compounds containing anions of negligible basicity (such as the conjugate bases of strong acids), solubility is unaffected by changes in pH.

At 25 °C, the value of K_{sp} for BaF_2 (s) is 1.8×10^{-7}.

Calculate the molar solubility of BaF_2 (s) in 0.25 M NaF (aq) at 25 °C.

Write your answer in scientific notation using two significant figures.

	BaF_2 (s)	$\rightleftharpoons$	Ba^{2+} (aq)	+	$2F^-$ (aq)
Initial (atm)	-		0		0.25
Change (atm)	-		+x		+2x
Equil. (atm)	-		x		0.25 + 2x

$K_{sp} = 1.8 \times 10^{-7} = [Ba^{2+}][F^-]^2 = (x)(0.25 + 2x)^2$

Since K_{sp} is small, we can assume that 2x is much less than 0.25 (that is, $0.25 + 2x \approx 0.25$).

This gives us: $1.8 \times 10^{-7} = (x)(0.25)^2$

$x = 2.9 \times 10^{-6}$

A 150.0 mL sample of 0.0020 M $Ca(NO_3)_2$ (aq) is added to 150.0 mL of 0.060 M $NaIO_3$ (aq) at 25 °C.

Does a precipitate of $Ca(IO_3)_2$ (s) form? (The value of K_{sp} for $Ca(IO_3)_2$ (s) at 25 °C is 6.5×10^{-6}.)

Find $[Ca^{2+}]$ and $[IO_3^-]$ just after mixing:

- $[Ca^{2+}]\ after\ mixing = \frac{(0.0020\ mol/L)(0.1500\ L)}{(0.1500 + 0.1500)\ L} = 0.0010\ M$
- $[IO_3^-]^2\ after\ mixing = \frac{(0.060\ mol/L)(0.1500\ L)}{(0.1500 + 0.1500)\ L} = 0.030\ M$

Use these values to calculate Q_{sp}:

- $Q_{sp} = [Ca^{2+}][IO_3^-]^2 = (0.0010)(0.030)^2 = 9.0 \times 10^{-7}$ M

Since $Q_{sp} < K_{sp}$, the solution is undersaturated [too much reactant already compared to equilibrium] and $Ca(IO_3)_2$ (s) does not precipitate.

$Zn(OH)_2$ (s) ⇋ Zn^{2+} (aq) + $2OH^-$ (aq) $K_{sp} = 3 \times 10^{-7}$

A beaker contains a saturated solution of $Zn(OH)_2$ in distilled water, as represented by the equation above.

If the pH of the solution is decreased at constant temperature, which of the following will most likely occur?

✗ The value of K_{sp} will increase.

The value of the solubility-product constant depends only on temperature and not on pH. In this case, the temperature remains constant, so the value of K_{sp} will not change.

✓ The molar solubility of $Zn(OH)_2$ (s) will increase.

Decreasing the pH (adding H^+ ions) will decrease the concentration of OH^- in solution, causing the equilibrium position to shift to the right. As a result, the molar solubility of $Zn(OH)_2$ (s) will increase.

Unit 8: Acids and Bases

8.1: Introduction to Acids and Bases

[Water autoionization and Kw (article) | Khan Academy]

[The Hydronium Ion - Chemistry LibreTexts]

Remember that for any p-scale value (pH, pOH, pK_w, etc.), only the numbers to the right of the decimal point are significant. For example, if the pH value has two decimal places, the concentration should have two significant figures.

$K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C

- $[H_3O^+] = 10^{-pH}$

pH = -log$[H_3O^+]$, pOH = -log$[OH^-]$

-logK_w = pK_w = **pH + pOH = 14.00**

In a neutral solution, $[H_3O^+] = [OH^-]$

In an acidic solution, $[H_3O^+] > [OH^-]$

In a basic solution, $[H_3O^+] < [OH^-]$

Autoionization (self-ionization) of water: one water molecule acts as a Bronsted-Lowry acid and the other as a Bronsted-Lowry acid, resulting in the formation of hydronium (H_3O^+) and hydroxide (OH^-) in a 1:1 ratio. Thus, in pure water, the ratio of hydronium (H_3O^+) and hydroxide (OH^-) must be equal.

- K_w: the autoionization constant of water (temperature-dependent!)

As H^+ ions are formed, they bond with H_2O molecules in the solution to form H_3O^+ (the hydronium ion). This is because hydrogen ions do not exist in aqueous solutions, but take the form of the hydronium ion (H_3O^+) instead.

Although other kinds of dissolved ions have water molecules bound to them more or less tightly, the interaction between H^+ and H_2O is so strong that writing "H^+ (aq)" hardly does it justice, although it is formally correct. The formula H_3O^+ more adequately conveys the sense that it is both a molecule in its own right, and is also the conjugate acid of water.

$2H_2O \leftrightharpoons H_3O^+ (aq) + OH^- (aq)$

At 25 °C, the autoionization constant for water, K_w, has a value of 1×10^{-14}. At 0 °C, the value of K_w is less than 10×10^{-14}. Based on the information above, which of the following is true?

✗ **The value of ΔH° for the autoionization process is negative.**

According to the text, the value of K_w is lower at 0 °C than at 25 °C. Since the value of K_w decreases with temperature, the autoionization process must be endothermic (heat can be thought as a reactant), which means that the value of ΔH° for the process is *positive*, not negative.

✗ **The value of pK_w at 0 °C is less than 14.0.**

According to the text, the value of K_w is lower at 0 °C than at 25 °C. Since K_w and pK_w are inversely related, this means that the value of pK_w at 0 °C must be *greater*, not less, than 14.0 (the value of pK_w at 25 °C).

✓ **The pOH of pure water at 0 °C is greater than 7.0.**

According to the text, the value of K_w is lower at 0 °C than at 25 °C. Since a smaller K_w value favors the reactants more, this means that water must be ionized to a lesser degree at 0 °C than 25 °C. As a result, the concentration of OH^- is lower at 0 °C, resulting in a pOH that is greater than 7.0.

8.2: pH and pOH of Strong Acids and Bases

Strong acids (such as HCl, HBr, HI, HNO_3, $HClO_4$, and H_2SO_4) ionize completely in water to produce hydronium ions. The concentration of H_3O^+ in a strong acid solution is therefore equal to the initial concentration of the acid.

- For example, a solution of 0.1 M HNO_3 contains 0.1 M H_3O^+ and has a pH of 1.0.

Strong bases (such as Group 1 and 2 metal hydroxides) dissociate completely in water to produce hydroxide ions. The concentration of OH^- in a strong base solution can therefore be determined from the initial concentration of the base and the stoichiometry of the dissolution.

- For example, the strong base $Ba(OH)_2$ dissociates to give two OH^- ions per formula unit, so a 0.1 M $Ba(OH)_2$ solution has an OH^- concentration of 0.2 M.

8.3: Weak Acid and Base Equilibria

$K_a = \frac{[H_3O^+][A^-]}{[HA]}$

- K_a (weak acid)

$K_b = \frac{[HB^+][OH^-]}{[B]}$

- K_b (weak base)

$K_w = K_a \times K_b$ [quack= q + a + c + k]

$pK_a = -\log K_a$, $pKb = -\log K_b$

$HClO$ (aq) + H_2O (l) ⇋ H_3O^+ (aq) + ClO^- (aq)

The ionization of the weak acid HClO in water is represented by the equation above. What is the percent ionization of HClO in 0.20 M HClO (aq)? (For HClO (aq), $K_a = 4.0 \times 10^{-8}$.)

To calculate the percent ionization, we first need to determine the equilibrium concentration of H_3O^+. We can do so by preparing an ICE (Initial, Change, and Equilibrium) table in which x = $[H_3O^=]_{equil}$ and then solving for the value of x.

- Molar solubility (S): the molar concentration of the solid that dissolves/dissociates in water. Equal to 'x' in an ICE table. [do not need to take the denominator/reactants into consideration, because it's a solid and that will equal 1]

Let's start by setting up our ICE table:

	HClO (aq)	+	H_2O (l)	⇌	H_3O^+ (aq)	+	ClO^-
Initial (M)	0.20		–		0		0
Change (M)	-x		–		+x		+x
Equil. (M)	0.020 - x		–		x		x

Note that we ignored the small initial concentration of H_3O^+ resulting from the autoionization of water.

Next, let's find the value of x by substituting the equilibrium concentrations from our ICE table into the expression for K_a:

- $K_a = 4.0 \times 10^{-8} = [H_3O^+][ClO^-]/[HClO] = x^2/(0.020 - x)$

Since K_a is so small, we can make the approximation that x is much less than 0.020 (that is, $0.020 - x \approx 0.020$).

- $4.0 \times 10^{-8} = x^2/0.020$
- $x = [H_3O^+]_{equil} = 2.8 \times 10^{-5}$ M.

Now that we know the equilibrium concentration of H_3O^+, we can calculate the percent ionization of HClO in solution:

- % ionization = $[H_3O^+]_{equil}/[HA]_{init} \times 100\%$
- 2.8×10^{-5} M/0.020 M x 100% **= 0.14%**

Since the percent ionization is less than 5%, the approximation we made in the previous step is valid.

Acid	Concentration	pH
1	0.010 M	2.2
2	0.010 M	3.4
3	1.0 M	2.5
4	1.0 M	3.4

Which of the four monoprotic acids listed in the table above has the smallest K_a value?

The acid with the smallest K_a is the *weakest* acid. At a given concentration, a weaker acid forms a solution with a higher pH than a stronger acid does. Given this, we know that acid 2 must be weaker than acid 1, and acid 4 must be weaker than acid 3.

The solutions formed by acids 2 and 4 have the same pH (3.4) but different initial concentrations (0.010 M vs. 1.0 M).

To achieve a certain pH, a weaker acid will always require a higher initial concentration than a stronger acid. Given this, acid 4 must be weaker than acid 2.

So, the acid with the smallest K_a value is acid 4.

8.4: Factors Affecting Acid Strength

16.8: Molecular Structure and Acid-Base Behavior - Chemistry LibreTexts

The relative strength of an acid can be predicted based on its chemical structure.

In general, the stronger (more polar) the A–H or B–H$^+$ bond, the less likely the bond is to break to form H$^+$ ions and thus the less acidic the substance. This effect can be illustrated using the hydrogen halides:

Relative Acid Strength	HF	HCl	HBr	HI
H–X Bond Energy (kJ/mol)	570	432	366	298
pK_a	3.20	-6.1	-8.9	-9.3

The larger the atom to which H is bonded, the weaker the bond. As a result, acid strengths of binary hydrides increase as we go down a column or from left to right across a row of the periodic table.

- Binary hydrides: a class of compounds that consist of an element bonded to hydrogen, in which hydrogen acts as the more electronegative species.

Acidity is also greater when the H–A bond is weaker and when the conjugate base, A$^-$, is more stable.

8.5: Acid-Base Reactions

A 60. mL sample of 0.10 M HI (aq) is added to 40. mL of 0.20 M KOH (aq).

The pH of the resulting solution is in which range?

When HI (a strong acid) and KOH (a strong base) are mixed, the two species react quantitatively according to the following equation:

H_3O^+ (aq) + OH^- (aq) → $2H_2O$ (l)

We can estimate the pH of the resulting solution from the concentration of excess reactant (either H_3O^+ or OH^-) in the mixture.

Let's start by determining whether H_3O^+ or OH^- is in excess after the reaction:

	H_3O^+ (aq)	+	OH^- (aq)	→	$2H_2O$ (l)
Before reaction	0.060 L x 0.10 M = 0.0060 mol		0.040 L x 0.20 M = 0.0080 mol		–
After reaction	0.0060 - 0.060 = 0 mol		0.0080 - 0.0060 = 0.0020 mol		–

So, OH^- is in excess and will determine the pH of the solution.

Now, let's find the concentration of excess OH^- in the reaction mixture:

- $[OH^-]$ = mol OH^- in excess/L soln = 0.0020 mol/(0.060 + 0.040) L = 0.020 M

2.0×10^{-2} is between 1.0×10^{-1} and 1.0×10^{-2}

- The pOH must be between 1 and 2
- The pH of the resulting solution is **between 12 and 13**.

HCN (aq) + SO_4^{2-} (aq) ⇋ CN^- (aq) + HSO_4^- (aq) $K_{eq} < 1$

What is the strongest base in the reaction represented above?

HCN (aq)	+	SO_4^{2-} (aq)	⇌	CN^-	+	HSO_4^-
acid		base		conj. base		conj. acid

In an acid-base reaction, the equilibrium favors the side *opposite* the stronger acid and base. In this case, the equilibrium favors the reactants side ($K_{eq} < 1$), so the stronger base must be on the products side. So, the strongest base in the reaction is **CN^- (aq)**.

8.6: Buffers

Buffer solutions contain high concentrations of both a weak acid and its conjugate base (or a weak base and its conjugate acid). Because these components can neutralize added H^+ or OH^-, buffers are highly resistant to changes in pH.

Four ways to make a buffer:

- Partial neutralization of a weak acid with a strong base
- Partial neutralization of a weak base with a strong acid
- Weak acid + salt of conjugate base (ex: $HClO_2$ + $NaClO_2$)
- Weak base + salt of conjugate acid (ex: CH_3NH_2 + CH_3NH_3Cl)

Henderson-Hasselbalch equation: $pH = pK_a + \log\frac{[A^-]}{[HA]}$

- [HA] and [A$^-$] refer to the equilibrium concentrations of the conjugate acid-base pair used to create the buffer solution.
- When [HA] = [A$^-$], the solution pH is equal to the pK_a of the acid.

Adding which of the following to a 200. mL sample of 0.10 M HClO (aq) (pK_a = 7.40) would result in the formation of a buffer solution with a pH of 7.40?

✓ 100. mL of 0.10 M KOH (aq)

Adding 100. mL of 0.10 M KOH (aq) to a 200. mL sample of 0.10 M HClO (aq) would convert half of the HClO to ClO^-, resulting in a HClO/ClO^- buffer solution. Because the concentrations of HClO and ClO^- would be the same, the pH of the solution would be equal to 7.40.

✗ 200. mL of 0.10 M KOH (aq)

Adding 200. mL of 0.10 M KOH (aq) to a 200. mL sample of 0.10 M HClO (aq) would convert all of the HClO to ClO^-, resulting in a solution containing essentially only ClO^-. Since a buffer solution must contain high concentrations of *both* species in a conjugate acid-base pair, this solution would not be a buffer solution.

✗ 100 mL of 0.10 M KClO (aq)

It is true that adding 100. mL of 0.10 M KClO (aq) to a 200. mL sample of 0.10 M HClO (aq) would result in a HClO/ClO^- buffer solution. However, because the concentration of HClO would be twice that of ClO^-, the pH of the solution would be *less than*, not equal to, 7.40.

8.7: Acid-Base Titrations

The results of an acid–base titration can be summarized using a titration curve, which plots pH vs. volume of titrant added.

- Buffer region: the section of curve between the initial point and the equivalence point.
- Equivalence point [see: 4.7]: the **titrant-analyte mole ratio is the same in the balanced equation and the reaction**. The reactants are completely consumed. Indicates the end of the reaction.

For the titration of a strong acid with a strong base, the curve begins acidic and then turns basic after the equivalence point, which occurs at pH = 7.

For the titration of a strong base with a strong acid, the curve is similar except that it begins basic and turns acidic.

For the titration of a weak acid with a strong base, the pH curve is initially acidic and has a basic equivalence point (pH > 7). At the half-equivalence point, the concentrations of the buffer components are equal, resulting in **$pH = pK_a$**.

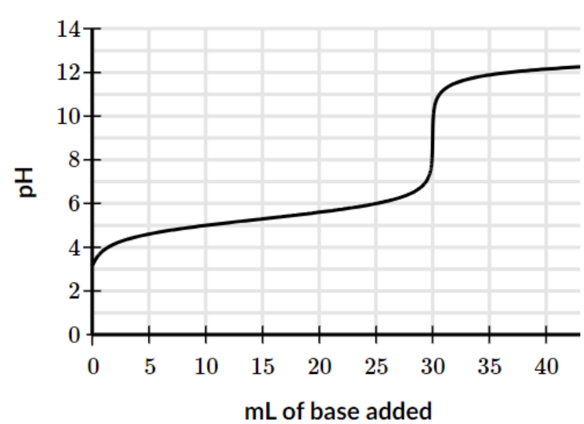
For the titration of a weak base with a strong acid, the pH curve is initially basic and has an acidic equivalence point (pH < 7). At the half-equivalence point, the concentrations of the buffer components are equal, resulting in **$pH = pK_a$** (where pK_a refers to the conjugate acid of the weak base).

To be useful, an indicator should change color at or near the equivalence point of a titration. For this to occur, the indicator must have a pK_a value close to the pH at the equivalence point.

A sample of a weak acid, HA, is dissolved in distilled water and titrated with a solution of strong base. The pH curve for the titration is shown in the graph to the right.

Using the curve, estimate the pK_a value for the acid.

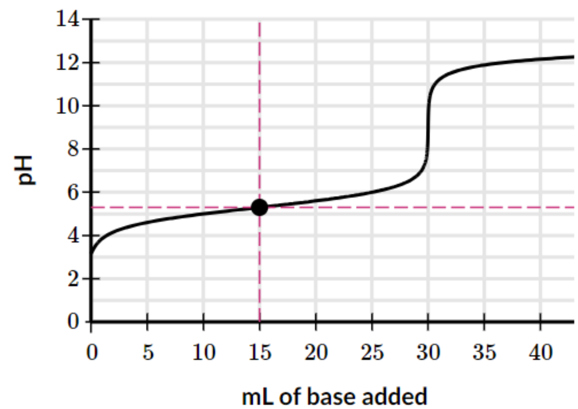
Write your answer to the tenths place.



We can find the pK_a value for the acid by estimating the pH at the half-equivalence point of the titration. At this point, the concentrations of HA and A^- are equal and pH = pK_a (Henderson-Hasselbalch equation).

Let's start by identifying the half-equivalence point on the titration curve. According to the graph, the equivalence point is at 30 mL (as evidenced by the sharp increase in pH at this volume). So, the half-equivalence point must be at 15 mL.

Next, let's estimate the pH at this point. The graph shows that the half-equivalence point is located above pH = 5.0 and below pH = 5.5, so a reasonable estimate for the pH is **anywhere between 5.1 and 5.4**.



A student is tasked with titrating a 25 mL sample of a weak monoprotic acid with 0.50 M NaOH (aq). To determine the equivalence point of the titration, the student is given a choice of the indicators listed in the table below.

Indicator	pK_a
Methyl orange	3
Bromocresol green	5
p-Nitrophenol	7
Thymol blue	9

Which of the following identifies for the titration?

To be useful, an indicator should change color at or near the equivalence point of a titration. For this to occur, the indicator must have a pK_a value close to the pH at the equivalence point.

The pH at the equivalence point is expected to be basic (pH > 7) since a weak acid is being titrated with a strong base.

So, the best indicator for the titration is thymol blue (pK_a > 7).

Unit 9: Applications of Thermodynamics

9.1: Introduction to Entropy

Enthalpy (H): a state function that describes the total heat content of a system.

Entropy (S): a measure of the number of microstates available to a system.

- Or: a thermodynamic quantity representing the unavailability of a system's thermal energy for conversion into mechanical work, often interpreted as the degree of disorder or randomness in the system.
- Microstates: the number of different possible arrangements of molecular position and kinetic energy at a particular thermodynamic state.

The number of available microstates increases **when matter becomes more dispersed**, such as when a liquid changes into a gas or when a gas is expanded at constant temperature.

It also increases when **energy becomes more dispersed**, which occurs with an increase in temperature.

In general, the entropy change for a reaction is **negative** ($\Delta S^{o}_{rxn} < 0$) when the number of moles of gaseous products is **less than** the number of moles of gaseous reactants.

- $\Delta S^{o}_{rxn} = \Sigma S^{o}\,(products)\ -\ \Sigma S^{o}\,(reactants)$ [multiply the S° value for each substance by the stoichiometric coefficient]

9.2: Gibbs Free Energy and Thermodynamic Favorability

When raising e to a power, the result has the same number of significant figures as there are decimal places in the power. Consider $e^{1.0} = 3$. The power has one decimal place, so the result has one significant figure.

Gibbs free energy: indicates the thermodynamic favorability of a physical or chemical process. When $\Delta G° < 0$, the process is thermodynamically favored (**spontaneous**).

- $\Delta G^o_{rxn} = \Sigma G^o_f(products) - \Sigma G^o_f(reactants)^*$
- $\Delta G^o_{rxn} = \Delta H^o_{rxn} - T\Delta S^o_{rxn}$
- $\Delta G^o = -nFE^o$
- $\Delta G^o = -RTlnK$ where K is the equilibrium constant and R is the universal gas constant (8.314 J mol^{-1} K^{-1}) (see: 9.5)

*Multiply the $\Delta G°_f$ value of each substance by its stoichiometric coefficient!

*Just like with enthalpy of formation, pure elements in their natural state have a Gibbs free energy of formation of 0!

A reaction is said to be driven by enthalpy or entropy when either factor helps make $\Delta G°_{rxn}$ negative.

T is in KELVIN and can only be positive!!!

$2H_2S$ (g) + SO_2 (g) → 3S (s) + $2H_2O$ (g) $\quad$ $\Delta H°_{rxn} < 0$

Under what temperature conditions is the reaction represented above thermodynamically favored?

$\Delta G°_{rxn} = \Delta H°_{rxn} - T\Delta S°_{rxn}$

The sign of $\Delta H°_{rxn}$ is given (negative).

Because the number of moles of **gaseous products** is *less* than the number of moles of gaseous reactants (2 vs 3), $\Delta S°_{rxn}$ is expected to be negative.

Since $\Delta H°_{rxn}$ and $\Delta S°_{rxn}$ are both negative, $\Delta G°_{rxn}$ will be negative only at *lower* temperatures (when the $T\Delta S°_{rxn}$) term is *smaller* in magnitude than the $\Delta H°_{rxn}$ term).

9.3: Free Energy of Dissolution

Dissolution: the dissolving of one substance in a solvent.

If $\Delta G^o_{soln} < 0$, the dissolution is thermodynamically favorable and the substance would dissolve in water.

- ΔG^o_{soln} is the free energy of dissolution.

1. Break up the solid:
    - ΔH_1 is positive: energy is required to break the bonds holding the solid together.
    - ΔS_1 is positive: when the solid is broken up, the particles have a greater number of possible positions and thus microstates.
2. Preparation of the solvent:
    - ΔH_2 is positive: use energy to break the intermolecular forces holding the solvent particles and move them far apart enough to fit a solute particle.
    - ΔS_2 is positive: more positions for particles.
3. Solvation (hydration if the solvent is water) – the interaction of the solute and the solvent
    - ΔH_3 is negative: energy is released when the solute and solvent come together.
    - ΔS_3 is negative: the solvent and solute are attracted to each other, so the number of positions is decreased.

$\Delta H^o_{soln} = \Delta H_1 [+] + \Delta H_2 [+] + \Delta H_3 [-]$; so, the magnitude of ΔH_3 determines whether or not the ΔH^o_{soln} is positive or negative.

The same logic applies to ΔS^o_{soln}.

9.4: Thermodynamics vs. Kinetics

Thermodynamics tells us what *can* occur during a process (will it go?), while kinetics tell us what *actually* occurs (how long will it take?).

9.5: Free Energy and Equilibrium

$\Delta G^{o} = -RTlnK$ where K is the equilibrium constant and R is the universal gas constant (8.314 J mol^{-1} K^{-1}) (convert units!)

$4NO_2$ (g) + O_2 (g) ⇋ $2N_2O_5$ (g) K_1 = 8.3 x 10^{-6}

Based on the information above, is the reaction below thermodynamically favored or not and why.

$2N_2O_5$ (g) ⇋ $4NO_2$ (g) + O_2 (g) K_2 = ?

$K_2 = 1/K_1 = 1/(8.3 \times 10^{-6}) > 1$

Since $K_2 > 1$, lnK_2 is positive and ΔG^{o}_{rxn} is negative:

- $\Delta G^{o}_{rxn} = -RTlnK_2$

So, the resection is thermodynamically favored because $K_2 > 1$.

[Quack 🦆]

9.6: Coupled Reactions

A thermodynamically unfavored reaction can be driven by coupling it to a favored reaction through one or more shared intermediates. The sum of the two reactions yields an overall reaction that has a negative ΔG° value.

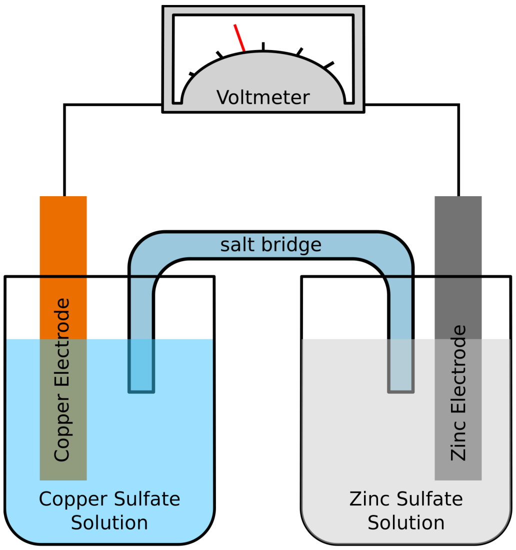
- Similar to Hess's law!

	Reaction	ΔG°
	$Cu_2S\ (s) \rightleftharpoons 2Cu\ (s) + \cancel{S\ (s)}$	$\Delta G^\circ = 86.1\ kJ/mol_{rxn}$
+	$\cancel{S\ (s)} + O_2\ (g) \rightleftharpoons + SO_2\ (g)$	$\Delta G^\circ = -300.4\ kJ/mol_{rxn}$
	$Cu_2S\ (s) + O_2\ (g) \rightleftharpoons 2Cu\ (s) + SO_2\ (g)$	$\Delta G^\circ = -214.3\ kJ/mol_{rxn}$

9.7: Galvanic (Voltaic) and Electrolytic Cells

Electrochemistry (article) | Khan Academy

Galvanic (voltaic) cells (video) | Khan Academy



Line notation: $Zn|Zn^{2+} || Cu^{2+}|Cu$

- anode [salt bridge] cathode

Galvanic (voltaic) cell: uses a thermodynamically favored redox reaction to generate an electric current.

Electrode: a conductor [anode or cathode for galvanic cells] that is used to make contact with a nonmetallic part of a circuit.

According to the mnemonic "Red Cat An Ox", reduction occurs at the cathode and oxidation occurs at the anode.

Salt bridge: ions flow through to keep the charge neutral.

Electrons flow from the anode to the cathode.

Cations flow to the cathode and anions flow to the anode to maintain charge balance.

In a spontaneous reaction (ΔG is negative):

- At the anode, the metal **loses electrons**, and metal ions enter the solution.
- At the cathode, the metal **gains electrons**, and metal ions leave the solution.

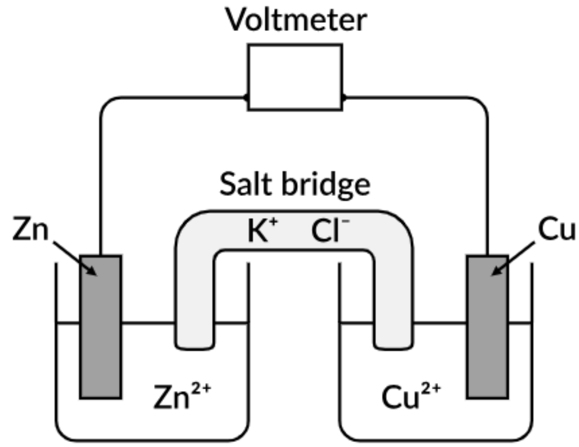
In an electrolytic reaction (ΔG is positive), it does the opposite.

$\Delta G^\circ = -nFE^\circ$: the equation for free energy

- ΔG°: standard free energy unit J
- n: moles of e^-
- F (Faraday): 1 F = 96458 C(oulomb) = 1 mol e^-

The diagram to the right shows a galvanic cell that has an electrode of Zn (s) immersed in a 1.0 M solution of $ZnSO_4$ (aq) and an electrode of Cu (s) immersed in a 1.0 M solution of $CuSO_4$ (aq). As the cell begins to operate, electrons flow from the Zn (s) electrode to the Cu (s) electrode.

What particle view best represents what occurs in the salt bridge as current flows through the cell? (Do not show solvent molecules.)

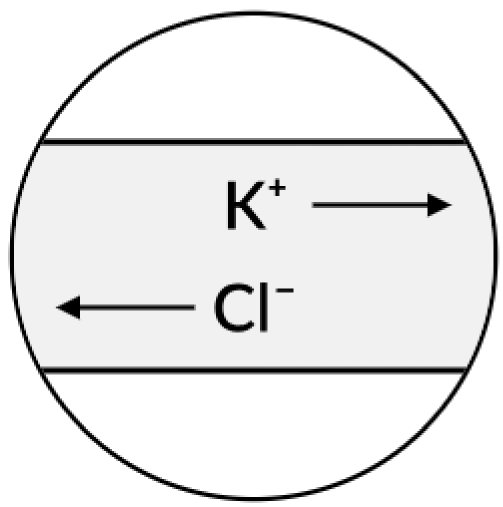


In an electrochemical cell, ions move within the salt bridge in a way that maintains charge balance in the two half-cells.

According to the text, electrons flow from the Zn (s) electrode to the Cu (s) electrode as the cell begins to operate. This will create a positive charge buildup in the zinc half-cell and a negative charge buildup in the copper half-cell.

To balance these charge buildups, the negative ions within the salt bridge will move toward the zinc half-cell (to the *left* in the particle view), and the positive ions will move towards the copper half-cell (to the *right* in the particle view).

So, the particle view that best represents what occurs in the salt bridge as current flows through the cell is:



A student sets up a galvanic cell that has a Pt (s) electrode and a AG (s) electrode. The half-reactions that occur at each electrode are represented in the table below.

Electrode	Half-Reaction
Pt (s)	Fe^{2+} (aq) → Fe^{3+} (aq) + e^-
Ag (s)	Ag^+ (aq) + e^- → Ag (s)

What happens to the mass of each electrode as the cell operates?

Pt (s) is not involved in the half-reaction that occurs at that electrode (Fe^{2+} is oxidized to Fe^{3+}), so the reaction does not form a solid and the mass of the Pt (s) electrode will not change as the cell operates.

In the half-reaction that occurs at the Ag (s) electrode, Ag^+ is reduced to Ag (s). Since Ag (s) is involved in the reaction, the mass of the Ag (s) electrode will change (in this case, increase, as the cell operates.

So, as the cell operates, the mass of the Ag (s) electrode increases, and the mass of the Pt (s) electrode remains the same.

9.8: Cell Potential and Free Energy

Cell potential: $E^{o}_{cell} = E^{o}_{red} + E^{o}_{ox}$ [calculated under standard conditions using 1.0 M at 25 °C]

- Standard cell potential = standard oxidation potential + standard reduction potential
- Unit: volt = Joules/coulomb of electrical charge.
- Use the reductions potential chart. Switch the sign for oxidation potential.
- **If the reaction is spontaneous**, the one with the higher reduction potential is more likely to be reduced. Since more positive is reduction, oxidation, the opposite of reduction, is more negative during a spontaneous reaction.
- **If the reaction is provided/not spontaneous**, flip the signs so the table entries can be added to find the full reaction.
- Even if the half-reactions are multiplied by an integer, the standard cell potential (E_{cell}^{o}) stays the same. This is because standard reduction potential is an intensive property and thus does not depend on the amount of material involved.

Ag^{+} (aq) + e^{-} → Ag (s) E^{o}_{red} = +0.80 V

Zn^{2+} (aq) + $2e^{-}$ → Zn (s) E^{o}_{red} = -0.76 V

Based on the standard reduction potentials given above, what is the value of E° for the galvanic cell?

The greater the E^{o}_{red} (standard reduction potential), the more likely it is to be reduced in a spontaneous reaction.

- Since 0.80 is greater than -0.76, Ag is reduced and Zn is oxidized.

Thus, the reaction is Zn (s) + $2Ag^{+}$ (aq) → Zn^{2+} (aq) + 2Ag (s).

- Zn (s) → Zn^{2+} (aq) + $2e^{-}$ E^{o}_{red} = 0.76 V

$E^{o}_{cell} = E^{o}_{red} + E^{o}_{ox}$ = 0.80 V + 0.76 V = 1.56 V

9.9: Cell Potential Under Nonstandard Conditions

Nernst Equation - Chemistry LibreTexts

Nernst equation: $E = E^{o} - \frac{RT}{nF} lnQ$ [cell potential not at 1.0 M and 25 °C]

If the temperature is at 25 °C: $E_{cell} = E^{0}_{cell} - \frac{0.0592}{n} logQ$

- E = reduction potential
- E° = standard potential
- R = the universal gas constant [8.314 J mol^{-1} K^{-1}]
- T= temperature in Kelvin
- n = ion charge (moles of electrons)
- F = Faraday's constant [96,485 C(oulombs)/1 mol e^{-}]
- Q = reaction quotient ($\frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}}$) **[solids and liquids have a value of 1]**

The equilibrium constant K_{eq} is proportional to the standard potential of the reaction.

- K > 1, E° > 0, reaction favors products formation (spontaneous!).
- K < 1, E° < 0, reaction favors reactants formation.

If ΔG° is negative, E° is positive and the reaction is spontaneous.

- Galvanic cells are spontaneous, electrolytic cells are not.

9.10: Electrolysis and Faraday's Law

Electrolysis equation: I = Q/t

- I = current (amperes) (1 amp = 1 coulomb/sec)
- Q = charge (coulombs)
- t = time (seconds)

In the electroplating process, 97.5 g of platinum metal is deposited from a solution of Pt^{4+} (aq) ions. How many coulombs of charge are transferred during this process?

$$97.5\ g\ Pt \times \frac{1\ mol\ Pt}{195\ gPt} \times \frac{4\ mol\ e^-}{1\ mol\ Pt} \times \frac{96{,}500\ C}{1\ mol\ e^-} = 193{,}000\ C$$